

A theory of memory for items and associations

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The authors dedicate this paper to the late Professor Bennet Murdock

Draft, May 19, 2026

Author Note

The authors dedicate this paper to the late Professor Bennet Bronson Murdock. May his memory be a blessing for all who knew him. We also gratefully acknowledge support from National Institutes of Health Grant MH55687. Thanks are due to Masaki Hori for programming Experiment 3 and for analyzing data from that experiment. Jeremy Caplan, Marc Howard, and Dan Rizzuto provided many valuable insights into the analysis of associative memory and successive memory tests. The code for all simulations and for the analysis of Experiment 1 and the data from Experiment 1 are available at <https://github.com/pennmem/CMR-IA>. Correspondence concerning this article should be addressed to Michael Jacob Kahana, kahana@psych.upenn.edu.

Abstract

We present a retrieved-context theory of memory for items, associations, and their interaction (CMR-IA). Our theory assumes an evolving representation of temporal context that binds to items and associations, allowing the rememberer to make judgments based on the occurrence of a mnemonic target within a particular context. In addition to the assumptions inherited from prior retrieved-context theories, CMR-IA assumes a conjunctive (Gestalt) representation for paired associates, increased attention to rare items, and variable thresholds for recognition decisions. We apply CMR-IA to key findings concerning recognition of items and associations, including effects of recency, similarity, receiver-operating characteristic curves, word frequency, differential forgettings of items and associations, and contiguity effects for successive probes. We also apply CMR-IA to cued recall phenomena, including serial position effects, distribution of correct responses and errors, contiguity effects, associative symmetry, and similarity effects. Finally, we ask whether CMR-IA can account for the dependencies between successive tests of item and associative memory. We show that combining a Gestalt associative mechanism with retrieved-context theory provides a good account for many empirical phenomena concerning item and associative memory. The analysis of successive memory tests highlights the important role of output encoding in our model.

Keywords: recognition, recall, association, context

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Studies of episodic memory have distinguished item-specific information and associative (or relational) information (Humphreys, 1978; Murdock, 1974). Item-specific information represents the encoding in memory of the attributes of an item occurring in a particular context. In contrast, associative information represents the encoding in memory of the relation between two co-occurring items. Associative information underlies the ability to learn that particular words appeared together in a study list, and item-specific information underlies the ability to recognize an item as having appeared in a study list. Although recognition is most often used to probe memory for item information, one can test recognition memory for associations by asking subjects to discriminate between correctly and incorrectly paired items (e.g., Hockley, 1992). The present paper aims to provide a unified theoretical account of memory for items and associations within the framework of retrieved-context theory. We apply our model to classic data regarding memory for items and associations. Further, we report and simulate new data on the interaction between item and associative memory, evaluated using the method of successive tests (Kahana, 2000; Kahana, Rizzuto, & Schneider, 2005).

Items and Associations

A number of experimental variables differentially influence item and associative memory. For example, people forget relational information more slowly than item information (Hockley, 1992). Whereas people exhibit superior recognition performance for rare as compared with common words, this effect does not appear for pairs of rare words as compared with pairs of common words (Clark, 1992), and in some cases, this effect even reverses (Lohnas & Kahana, 2013).

Whereas the distinction between item and associative information pertains to the representation being stored and retrieved from memory, the distinction between recognition and recall pertains to how the information is retrieved. In a recognition task, subjects typically judge whether a target item or association occurred in the context of a

just-presented list. In a recall task, subjects attempt to recall a target item based on a specific associated cue item. Both cued recall and associative recognition measure memory for associative information, but cued recall involves a process that leads to the production of an item, sometimes referred to as deblurring. Associative recognition, in contrast, involves a decision process that leads subjects to convert an index of retrieved information to a “yes” versus “no” response to the question of whether the association occurred in the target list.

Context-Maintenance and Retrieval Model for Items and Associations (CMR-IA)

We develop a model of memory for item and associative information that builds on a class of models that conceive of episodic memory as the interaction between content and context information. Content information is simply the information experienced at a given time, and context is a slowly drifting representation of background features that evolve due to our experiences and the associations they evoke (Manning, 2024). Experiencing or recalling a memory reinstates its earlier context(s), which in turn updates the present state of context and associates with subsequent experiences. The recursive nature of context allows associations from the distant past to inform new learning and binds items never experienced together (Cohen & Kahana, 2022; Wachter & Kahana, 2024).

Models based on retrieved-context offer a rigorous account for phenomena observed in studies of free recall where the cue for retrieval is context itself (Lohnas, Polyn, & Kahana, 2015; Healey & Kahana, 2016). Recent work has extended this class of models to account for sequence production and serial recall (Logan, 2021; Logan & Cox, 2023; Lohnas, 2025; but also see Osth & Hurlstone, 2023 for criticism). Here, we ask whether a similar context-based theoretical approach can help elucidate recognition and cued recall of item and associative information. As our model extends the Context-Maintenance and Retrieval (CMR) family to item and associative memory, we refer to it as CMR-IA.

Our paper is organized as follows. Section 1 presents a selective review of early

models of item and associative memory, providing a theoretical backdrop to our work (Osth & Dennis, 2024, provide a comprehensive review of the recognition modeling literature). Section 2 presents CMR-IA. Section 3 evaluates CMR-IA against key findings concerning item and associative memory. Section 4 presents new data on successive tests of item and associative information. Section 5 evaluates CMR-IA in relation to these new data. Section 6 contains a comprehensive general discussion and conclusion.

Theoretical Backdrop

Strength theory provided the earliest account of item and associative information and their relations (Wickelgren & Norman, 1966). According to this view, studying items on a list strengthens associations between each list item and some representation of the list itself. Outcomes of item recognition, associative recognition, and cued recall tests depend on the strength of these associations. Consistent with the observation that item recognition is usually easier than cued recall, strength theory held that the essential difference between these tasks is that item recognition could be successfully performed with weaker associations than either cued recall or associative recognition (Kahana, 2012).

Researchers found strength theory's account of the relation between item recognition and cued recall to be deficient, as many experimental variables have opposing effects on these tasks. Subjects easily recognize rare as opposed to common words as having occurred on a recent list, but they find rare words more difficult to recall (Gregg, 1976; Kinsbourne & George, 1974; MacLeod & Kampe, 1996; Lohnas & Kahana, 2013). Other dissociations include the effects of intentionality on encoding (Glenberg & Bradley, 1979; R. M. Schwartz & Humphreys, 1974), the effects of context-change on retrieval, and the effects of medial-temporal lobe damage on recognition of dissimilar items (Holdstock et al., 2002).

Generate-recognize theory quickly emerged as a preferred alternative account (e.g., Bahrck, 1970; Anderson & Bower, 1972). This theory posited that recall involves two stages: Subjects first generate possible responses, and then apply a recognition test to decide whether any of the generated responses were on the list. The recognition task lacks

a generation stage. A strong version of generate-recognize theory predicts that recallable items will always be recognized, contrary to many studies (e.g. Tulving, 1968; Tulving & Thompson, 1973; Flexser & Tulving, 1978).

A new generation of simulation-based, mathematical models of recognition and recall memory emerged in the 1970s and 1980s. These models often assumed distributed representations of items and associations, and operations that simulated the processes of encoding and retrieval (Kahana, 2020; Cox & Shiffrin, 2024). To illustrate how these models simulate data from recognition and recall, let us consider an experiment in which subjects first study a list of randomly paired common words, denoted $\mathbf{f}_i - \mathbf{g}_i$. Later, during a test phase, the experimenter may assess memory in one of several ways: by presenting a single word and asking whether it appeared in any of the studied pairs (item recognition), by presenting a pair of words and asking whether both items in the test pair appeared together as one of the studied pairs (associative recognition), or by presenting a single word from one of the studied pairs and asking subjects recall its mate (cued recall).

These models represent an item in memory as an N -dimensional column vector, so that column vector pair $(\mathbf{f}_i, \mathbf{g}_i)$ represents the i -th pair of studied items. These paired vectors combine in memory M as follows:

$$M = \sum_{i=1}^L (\mathbf{f}_i + \mathbf{g}_i)(\mathbf{f}_i + \mathbf{g}_i)^\top,$$

that is, the model forms a sum of the two item vectors in each pair and then multiplies the sum by its transpose to form an outer-product matrix. Rather than storing each matrix separately in memory, the model posits that they are combined by simple summation. In this manner, a single memory matrix represents all of the studied pairs in a given list (for the moment, assume that memory begins the experiment as a tabula rasa; following study, it only contains pairs from the encoded list). Kahana et al. (2005) discussed these assumptions and generalized this approach to convolution-correlation models.

In this model, cued recall operates by premultiplying the memory matrix, M , with the cue item, $\mathbf{f}_k$. This yields an approximate representation of the studied pair, $\mathbf{f}_k + \mathbf{g}_k$, as given by the following equation:

$$\begin{aligned}
 M\mathbf{f}_k &= \sum_{i=1}^L (\mathbf{f}_i \mathbf{f}_i^\top + \mathbf{f}_i \mathbf{g}_i^\top + \mathbf{g}_i \mathbf{f}_i^\top + \mathbf{g}_i \mathbf{g}_i^\top) \mathbf{f}_k \\
 &= (\mathbf{f}_k + \mathbf{g}_k)(\mathbf{f}_k \cdot \mathbf{f}_k + \mathbf{g}_k \cdot \mathbf{f}_k) \\
 &\quad + \sum_{l \neq k} (\mathbf{f}_l + \mathbf{g}_l)(\mathbf{f}_l \cdot \mathbf{f}_k + \mathbf{g}_l \cdot \mathbf{f}_k) \\
 &= \mathbf{f}_k + \mathbf{g}_k + \text{error}
 \end{aligned}$$

The match between the retrieved trace and the cue item, given by $M\mathbf{f}_k \cdot \mathbf{f}_k$, then determines the probability of successful cued recall. If the vectors are orthonormal, retrieval is error-free. Because vector addition is commutative, this implementation of cued recall is inherently symmetrical. Even if the strengths of the $\mathbf{f}$ and $\mathbf{g}$ items are differentially weighted, the forward and backward hetero-associative terms would have the same strength. The symmetric nature of these linear models also applies to their nonlinear variants (Hopfield, 1982).

Recognition operates by first cueing memory with $\mathbf{g}_i$ and retrieving its associated pattern, as in the cued recall model described above. The similarity between this retrieved vector and the cue determines whether the subject recognizes the association. Using the dot-product as our measure of similarity, as in Murdock's Theory of Distributed Associative Memory (TODAM; Murdock, 1982), we obtain the following expression:

$$M\mathbf{g}_i \cdot \mathbf{g}_i = (\mathbf{f}_i + \mathbf{g}_i) \cdot \mathbf{g}_i + \sum_{k \neq i} (\mathbf{f}_k + \mathbf{g}_k) \cdot \mathbf{g}_i$$

If the resulting similarity, given by the above expression, exceeds a threshold, the model will generate a "yes" response. Or, we can use this expression as the input to a stochastic decision process, which would allow us to obtain predictions regarding response times to various cue types (Hockley & Murdock, 1987).

Summed-similarity and signal-detection models of recognition memory, under certain assumptions, constitute special cases of the matrix model. This model also provides a natural account for interference effects in associative memory, and it predicts greater correlations between cued recall and associative recognition than between cued recall and item recognition. Kahana et al. (2005) provided a mathematical analysis of this model and comparisons with several other classes of distributed memory models, including those that use vector convolution as the associative operator (Murdock, 1982; Metcalfe-Eich, 1982; Metcalfe, 1985). By adding some basic auxiliary assumptions, such as probabilistic encoding (Murdock & Lamon, 1988; Murdock & Kahana, 1993a) and output encoding, Kahana et al. (2005) showed how each of these classic models can account for a range of recognition and recall phenomena, including the correlations between successive recognition and recall tests as seen in the Tulving-Wiseman relation (Kahana, 2000).

A key limitation, however, of the aforementioned models is the tabula rasa assumption. By resetting memory at the start of each list, these models do not need to grapple with interactions between memories learned in distinct temporal contexts, such as the different lists of a multi-trial experiment. To address this problem, a number of theories introduce the concept of a list context that associates with each presented item and pair (Humphreys, Pike, Bain, & Tehan, 1989; Osth & Dennis, 2015; Osth, Jansson, Dennis, & Heathcote, 2018; Cox, 2024). However, models that include a dynamical representation of spatiotemporal context subsume a broader array of memory phenomena, including the effects of recency and contiguity seen at multiple time scales (see Kahana, 2020, for a review).

2. CMR-IA: A Context Maintenance and Retrieval Model for Item and Associative Memory

Here we develop CMR-IA, a model of memory for item and associative information based upon the ideas of retrieved-context theory embodied in Polyn, Norman, and Kahana (2009); Lohnas et al. (2015); Healey and Kahana (2016); Cohen and Kahana (2022);

Wachter and Kahana (2024). In our usage, context refers to a mathematical representation of a psychological construct. At the psychological level, we wish to capture the idea that people form associations between items and their situational, temporal, and/or spatial context (Carr, 1931; McGeoch, 1932; Hollingsworth, 1928; Robinson, 1932; Underwood, 1945). This idea finds its clearest early expression in Bower’s (1972) multi-attribute temporal context model. According to Bower’s model, contextual representations comprise a multitude of fluctuating features, defining a vector that slowly drifts through a multidimensional context space. These contextual features form part of each memory, combining with other aspects of externally and internally generated experience. Because a unique context vector marks each remembered experience, and because context gradually drifts, the context vector conveys information about the time in which an event was experienced. Bower’s model, which drew heavily on the classic stimulus-sampling theory developed by (Estes, 1955), placed the ideas of temporal coding and internally generated context on a sound theoretical footing, and provided the basis for the retrieved context theory developed by Howard and Kahana (1999, 2002).

By allowing for a dynamic representation of temporal context, items studied on a given list will share more of their contextual attributes than items studied on different lists (Bower, 1972). If the contextual change between lists is sufficiently great, and if the context at time of test is similar to the context encoded at the time of study, then recognition memory judgments should largely reflect the presence or absence of the probe (test) item within the most recent (target) list, rather than the presence or absence of the probe item on earlier lists. This enables a multitrace summed-attribute similarity model to account for many of the major findings concerning recognition memory and other types of memory judgments.

To model contextual drift, we define a multidimensional context vector, $\mathbf{c} = (\mathbf{c}(1), \mathbf{c}(2), \dots, \mathbf{c}(N))$, and specify a process for its temporal evolution. For example, we could specify a unique random set of context features for each list, or for each experience

encountered in a particular situational context. However, contextual attributes fluctuate due to many internal and external variables that vary at many different time scales. To capture this idea, Murdock (1997) posited an autoregressive model for contextual drift, $\mathbf{c}_i = \rho \mathbf{c}_{i-1} + \sqrt{1 - \rho^2} \epsilon$, where ϵ is a random vector whose elements are each drawn from a Gaussian distribution, and where i indexes each item presentation. The variance of the Gaussian is defined such that the inner product $\epsilon_i \cdot \epsilon_j$ equals one for $i = j$ and zero for $i \neq j$. Accordingly, the similarity between the context vector at time steps i and j falls off exponentially with the separation: $\mathbf{c}_i \cdot \mathbf{c}_j = \rho^{|i-j|}$. Thus, the change in context between the study of an item and its later test increases with the number of items intervening between study and test, producing the classic forgetting curve (Kahana, Diamond, & Aka, 2024).

Glenberg and Swanson (1986) suggested that an evolving context representation could explain the phenomena of long-term recency, and Mensink and Raaijmakers (1988) demonstrated how a variable context signal in the SAM model could account for the major results obtained in studies of associative interference. Whereas these earlier formulations saw context as a cue for items, they did not explicitly consider items as a cue for context. That is, drifting contextual representations changed the accessibility of items, but item retrieval did not systematically alter the state of context.

Retrieved-context theory (RCT; Howard & Kahana, 1999, 2002) introduced the idea that remembering an item calls back its encoding context, which in turn serves as a retrieval cue for subsequent recalls. At its most basic level, this theory proposes that items and context become reciprocally associated during study: Context retrieves items, and items retrieve context. As the learner encodes or retrieves an item, RCT assumes that new associations form both between the item’s feature representation and the current state of context (stored in the matrix M^{FC}) and between the current state of context and the item’s feature representation (stored in the matrix M^{CF}). These associations follow a

Hebbian outer-product learning rule, as in earlier models of associative memory:

$$\begin{aligned} M_t^{FC} &= M_{t-1}^{FC} + \gamma_{FC} \mathbf{c}_i \mathbf{f}_i^\top \\ M_t^{CF} &= M_{t-1}^{CF} + \gamma_{CF} \mathbf{f}_i \mathbf{c}_i^\top, \end{aligned} \quad (1)$$

where γ_{FC} and γ_{CF} parameters determine the learning rates for item-to-context and context-to-item associations, respectively. Through these associative matrices, the presentation or recall of an item evokes its prior associated contexts, and cueing with context retrieves items experienced in overlapping contextual states.

Howard and Kahana (2002) further suggested that the mechanism of retrieved context was the driver of context evolution itself. Rather than context evolving randomly, they proposed that the currently active item, $\mathbf{f}_i$, retrieves (a weighted sum of) its previous contextual states, via the M^{FC} matrix, which in turn serves as the input to the context evolution equation. Normalizing this input yields the recursion:

$$\begin{aligned} \mathbf{c}_i^{\text{IN}} &= \frac{M^{FC} \mathbf{f}_i}{\|M^{FC} \mathbf{f}_i\|} \\ \mathbf{c}_i &= \rho_i \mathbf{c}_{i-1} + \beta \mathbf{c}_i^{\text{IN}} \\ \rho_i &= \sqrt{1 + \beta^2 [(\mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}})^2 - 1]} - \beta (\mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}}), \end{aligned} \quad (2)$$

where ρ_i is a constant ensuring that $\|\mathbf{c}_i\| = 1$, and where model parameter β governs the rate of contextual drift (a large value of β causes context states to decay quickly). During list recall, or when items are repeated, context recursively updates based on the past contextual states associated with the recalled or repeated item¹. Because context is always

¹ Prior to proposing the recursion in Equation 2, Howard and Kahana (2002) considered the simpler idea that items themselves form the input to context, as given by $\mathbf{c}^{\text{IN}} = \mathbf{f}_i$. Although this formulation had several desirable properties, it failed to capture the idea that context includes more than the perceptual and conceptual features of an item, but rather embodies its history of prior associated states. For example, when studying the word “chocolate”, one would be expected to update context with its past associated mental representations. Similarly, when recalling “chocolate”, both the encoding context of the study list and the past associations should come to mind.

of unit length, it can be thought of as a point on a hypersphere, with β determining the distance it travels with the newly presented item and $M^{FC}\mathbf{f}_i$ determining the direction of travel.

In modeling free recall, we use the current state of context $\mathbf{c}_t$ as a cue to retrieve items, via associations stored in M^{CF} :

$$\mathbf{f}_t^{\text{IN}} = M^{CF}\mathbf{c}_t. \quad (3)$$

The resulting $\mathbf{f}_t^{\text{IN}}$ gives the degree of support, or activation, for each item in the model’s lexicon. Different implementations of RCT have employed various models to simulate the recall process. Several early papers (e.g., Howard & Kahana, 1999) used a probabilistic choice rule similar to the sampling rule in SAM, but this approach requires additional assumptions about recall termination. Sederberg, Howard, and Kahana (2008) modeled the recall process as a competition among racing accumulators (one for each candidate recall), based on the decision model proposed by Usher and McClelland (2001). In this model, the first word whose accumulator crosses its threshold, prior to the end of the recall period, wins the competition. The contextual retrieval process in RCT produces the temporal and semantic organization seen in the dynamics of memory retrieval. Readers will find a more detailed review of these modeling assumptions in Kahana (2020) and Manning (2024).

Data on the dynamics of free recall have provided a rich testbed for the development of retrieved-context theory (Kahana, 2020; Lohnas, 2024). The cue-dependent nature of sequential retrievals in free recall lends itself particularly well to this type of theory. In recent work, Logan (2021; 2023) has applied a very similar modeling framework to explain data on various types of serial recall tasks. Here, we evaluate whether a retrieved-context theory imbued with a Gestalt model of associations can help us understand memory for items, associations, and their interactions.

Using Context Similarity to Model Item Recognition

A recognition test probe will retrieve its associated context state in the same way that items retrieve their associated contexts during encoding and retrieval. By comparing this retrieved context to the current state of context, we can assess the degree of similarity between the current context and the encoding context of the item. This comparison informs the recognition decision (Healey & Kahana, 2016). Our model judges an item $\mathbf{f}_i$ as “old” when the similarity between the context retrieved by $\mathbf{f}_i$ ($\mathbf{c}_i^{\text{IN}}$) and the current context $\mathbf{c}_t$ exceeds the item recognition threshold c_{thresh}^{item} :

$$\begin{aligned}\mathbf{c}_i^{\text{IN}} &= \frac{M^{FC} \mathbf{f}_i}{\|M^{FC} \mathbf{f}_i\|} \\ \mathbf{c}_i^{\text{IN}} \cdot \mathbf{c}_t &> c_{thresh}^{item}.\end{aligned}\tag{4}$$

After making a recognition judgment, our model integrates $\mathbf{c}_i^{\text{IN}}$ into the current state of context using Equation 2. Healey and Kahana (2016) assumed that context updating did not occur for “new” judgments. Here we generalize their approach by assuming that context updating takes place for both “old” and “new” judgments (in the simulations below, we assume the same rate of contextual drift for both types of judgments, but this could be generalized if needed). Finally, for items judged as “old”, the model also updates M^{FC} and M^{CF} following Equation 1, encoding the item with the current state of context. This mechanism plays an essential role in capturing correlations between successive memory tests as discussed in Section 5.

An extensive literature on recognition memory has now established that subjects can dynamically adjust their recognition thresholds within lists of a memory experiment, based on the expected distribution of similarities between old and new items (Hirshman, 1995; Rhodes & Jacoby, 2007; Han & Dobbins, 2008). We thus allow c_{thresh}^{item} to differ across items due to varying expected context similarities, as described further below.

Consider the scenario in which subjects studied a list of unrelated items. We would

model these items as orthonormal basis vectors in feature space. In this case, the following summation defines the similarity between the current context and the context retrieved by $\mathbf{f}_i$ under a simplified version of CMR:

$$\sum_j \mathbf{c}_j \mathbf{f}_j^\top \mathbf{f}_i \cdot \mathbf{c}_t = \sum_j (\mathbf{c}_j \cdot \mathbf{c}_t) (\mathbf{f}_j \cdot \mathbf{f}_i) = \mathbf{c}_i \cdot \mathbf{c}_t.$$

For the case of a list with a single repeated element occurring in positions i and k , we would have two matching context terms, $\mathbf{c}_i$ and $\mathbf{c}_k$, but the state of $\mathbf{c}_t$ will also be altered due to the context evolution equation. Specifically, $\mathbf{c}_t$, in this case, will be more similar to the earlier of the two retrieved contexts than it would in a list without repetitions, thus producing a greater increase in similarity than would be expected on the basis of the repetitions themselves (see Siegel & Kahana, 2014). The case of similar items should fall between that of repeated and unrelated items.

Modeling Associative Recognition and Cued Recall

Extending CMR to associative recognition and cued recall requires us to posit a mechanism for associative learning. Here, two logical possibilities emerge: 1) Treat the association as a list of two items, each associated with its own context, but allow context to drift more between pairs than within pairs. This approach aligns with the associative isolation hypothesis proposed by Caplan (2005). 2) Treat the pair as a single item that combines features of the two constituent elements. In this case, the model would essentially create a new feature vector for the pair, unitizing the elements into a single memory associated with a single context. This unitization approach accords with the Gestalt conceptions of associative learning and the findings of associative symmetry (Kahana, 2002).

Although logically distinguishable, these two approaches are by no means mutually exclusive. One could imagine that sometimes subjects construct a new item from two constituents, while at other times they treat the pair as if it were a two-item list.

Furthermore, although studies on cued recall generally find symmetric retrieval of forward and backward associations, associative recognition might not exhibit this symmetry (Yang et al., 2013, and Experiment 2 of this paper). Finally, subjects can remember the order of a pair of studied items, suggesting that temporal structure does find its way into the memorial representation of a pair. We adopt the second hypothesis here for the sake of parsimony, but we remain open to both possibilities.

In modeling associative memory, CMR-IA sums the vector representations of the paired associates $\mathbf{f}_i = \mathbf{f}_{i_1} + \mathbf{f}_{i_2}$ to construct a composite memory. It then updates context using Equation 2 and updates M^{FC} and M^{CF} using Equation 1. As in prior work, CMR-IA initializes M^{FC} and M^{CF} with pre-experimental semantic associations and applies a primacy gradient when updating M^{CF} (Sederberg et al., 2008; Polyn et al., 2009; Lohnas et al., 2015; Healey & Kahana, 2016). See Appendix A for more details.

The retrieval process for associative recognition and cued recall begins with contextual retrieval. In associative recognition, CMR-IA first updates the context using one of the items in the test pair. For example, when presented with the pair, $(\mathbf{f}_{i_1}, \mathbf{f}_{i_2})$, CMR-IA updates the current context with $\mathbf{f}_{i_1}$ through Equation 2, denoting the updated context as $\mathbf{c}'_t$. The model then compares this context with the context retrieved by $\mathbf{f}_{i_2}$, represented as $\mathbf{c}_{i_2}^{\text{IN}}$. CMR-IA responds “yes” if the similarity between $\mathbf{c}_{i_2}^{\text{IN}}$ and $\mathbf{c}'_t$ exceeds the threshold c_{thresh}^{assoc} , in a manner akin to Equation 4:

$$\begin{aligned} \mathbf{c}_{i_1}^{\text{IN}} &= \frac{M^{FC}\mathbf{f}_{i_1}}{\|M^{FC}\mathbf{f}_{i_1}\|}, \quad \mathbf{c}_{i_2}^{\text{IN}} = \frac{M^{FC}\mathbf{f}_{i_2}}{\|M^{FC}\mathbf{f}_{i_2}\|} \\ \mathbf{c}'_t &= \rho_t \mathbf{c}_t + \beta \mathbf{c}_{i_1}^{\text{IN}} \\ \mathbf{c}_{i_2}^{\text{IN}} \cdot \mathbf{c}'_t &> c_{thresh}^{assoc}. \end{aligned} \tag{5}$$

CMR-IA also integrates $\mathbf{c}_{i_2}^{\text{IN}}$ into the current context using Equation 2 after the recognition judgment.

For cued recall, CMR-IA first updates the current context with the cue $\mathbf{f}_i$ using

Equation 2. Then, the model utilizes the updated context $\mathbf{c}'_t$ to retrieve items from memory in the same manner as free recall (see Equation 3):

$$\begin{aligned}\mathbf{c}'_t &= \rho_t \mathbf{c}_t + \beta \frac{M^{FC} \mathbf{f}_i}{\|M^{FC} \mathbf{f}_i\|} \\ \mathbf{f}_t^{\text{IN}} &= M^{CF} \mathbf{c}'_t.\end{aligned}\tag{6}$$

If the model successfully retrieves an item, CMR-IA updates $\mathbf{c}'_t$ with the retrieved context using Equation 2. In other respects, CMR-IA inherits the assumptions of the CMR2 model developed by Lohnas et al. (2015) and Healey and Kahana (2016). See Appendix A for more details.

CMR-IA allows for the possibility that subjects will learn context-to-item associations and item-to-context associations for recognized items and pairs. Similarly, successful cued recall of an item can lead to the storage of item-to-context associations. Although output encoding has also been envisioned in prior CMR implementations, it has only rarely been introduced into simulations. Here, we use output encoding in all simulations involving repeated tests², where CMR-IA also updates M^{FC} and M^{CF} for recognized items, recognized pairs, or completed pairs in cued recall, following Equation 1. This mechanism plays an essential role in modeling the correlations between successive memory tasks.

3. Evaluating CMR-IA’s Account of Benchmark Phenomena

We evaluate CMR-IA’s ability to account for a set of key benchmark phenomena concerning memory for item and associative information. Our first set of simulations concerns data on item and associative recognition. Our second set of simulations concerns data on cued recall.

² Output encoding does not affect tests with no repeats, as long as we represent items as orthonormal basis vectors.

Simulations 1-4: Item and Associative Recognition

Simulation 1 examines the joint effects of recency and similarity on performance in a continuous recognition task based on the classic methods of Shepard and Teghtsoonian (1961) and Hockley (1982). Simulation 2 examines the contiguity effect in both item recognition and associative recognition, fitting the data presented in G. Schwartz, Howard, Jing, and Kahana (2005) and Osth and Fox (2019). Simulation 3 looks at the difference in forgetting rates between item recognition and associative recognition (Hockley, 1992). Simulation 4 examines the word frequency effect in item recognition (Lohnas & Kahana, 2013). Code for all of the simulations reported here may be obtained at <https://github.com/pennmem/CMR-IA>.

Experiment 1

To elucidate the joint effects of recency and similarity on recognition performance, we conducted a large-scale online study using a continuous item recognition procedure.

Methods

Subjects. We recruited 657 subjects from Amazon Mechanical Turk for the experiment. We excluded 109 subjects who failed to respond within the allotted time frame for more than 250 of the items.

This study was conducted as a demonstration experiment for students in a Human Memory course taught by M.J.K. at the University of Pennsylvania. The study was not pre-registered. Data and code for this experiment are available at <https://github.com/pennmem/CMR-IA>.

Procedure. Subjects completed a continuous recognition task (Shepard & Teghtsoonian, 1961). Rather than having separate study and test phases, participants provide recognition judgments on each item, indicating whether the item occurred previously in the list. To manipulate semantic similarity, we selected items from a pool of 300 words comprising 12 moderately-prototypical exemplars from each of 25 semantic categories (Weidemann et al., 2019; Kragel et al., 2017). Following standard

methods (Murdock, 1974), we only began repeating items after having first presented a buffer of once presented items (20 in our case). We then arranged the remaining 280 so that the first presentation of a word belonged to either a block of 2–6 items from the same category or a block of 1–6 items from different categories, leaving positions for repetitions in the overall sequence of 580 presentations. We then presented the repetitions of those items randomly within the unfilled positions. This procedure ensured variation in both temporal lag and semantic similarity among old words.

During each trial, subjects viewed a word for 1000 ms, followed by a 1000 ms inter-stimulus interval. Subjects placed their left and right hand on the following keys, one finger per key: “a”, “s”, “d”, “f”, “j”, “k”, “l”, “;”. Subjects pressed one of the eight keys in response to each probe, rating their confidence that they had seen it before from 8 (sure old) to 1 (sure new) (G. Schwartz et al., 2005).

Analysis

The study-test lags approximately follow an exponential distribution, so we partitioned them into five bins: Bin 1 (lag less than 8), Bin 2 (lag 8 to 20), Bin 3 (lag 21 to 54), Bin 4 (lag 55 to 148), and Bin 5 (lag greater than 148)³. Consistent with Hockley (1982), the hit rate decreases as the lag between study and test increases, as seen in Figure 1A. We defined items as having high semantic similarity when two or more same-category items appeared within the previous eight items; we defined items as having low semantic similarity when zero or one same-category item appeared within the previous eight items. For statistical inference, we fitted a Linear Mixed Model with study-test lag bin, similarity level, and their interactions as fixed effects. We used a maximal random effects structure that ensures non-singularity, following the procedure recommended by Bates, Kliegl, Vasishth, and Baayen (2018). The Satterthwaite-approximated ANOVA indicated that both recency, measured by study-test lag, and similarity significantly affect

³ Items with lags less than e^2 belong to Bin 1, items with lags greater than e^2 while less than e^3 belong to Bin 2, and so on.

hit rate (recency: $F(4, 804.82) = 85.72$, $p < .001$; similarity: $F(1, 246.34) = 5.71$, $p = .018$). The interaction between these factors does not show significance ($F(4, 522.11) = 0.95$, $p = .435$).

A decreasing hit rate does not necessarily imply degraded memory. If, for instance, as subjects tend to respond more conservatively with increasing study-test lag, one might observe hit rates decline along with false alarm rates. To address this, we calculated local false alarm rates (Murdock & Kahana, 1993b) for each study-test lag bin and similarity level. We checked each old item to determine if it had an adjacent new item. If it did, we considered this new item to belong to the same study-test lag bin as the old item⁴. We established the similarity level of the new item based on the number of same-category items within the previous eight items, as described above (Figure 1C). The ANOVA for local false alarm rate revealed significant effects of recency ($F(4, 1059.42) = 4.57$, $p = .001$) and similarity ($F(1, 504.97) = 21.15$, $p < .001$), but the interaction between these factors did not reach the $p < 0.05$ threshold ($F(4, 1020.39) = 2.10$, $p = .079$). Analyzing Figure 1A and Figure 1C together shows that greater recency improves recognition performance. However, higher similarity tends to produce both increased hit rates and increased false alarm rates, raising the question of whether similarity affects memory, decision criterion, or both.

We used the confidence ratings gathered in this experiment to calculate A_z , an estimate of the area under curve⁵ (Stanislaw & Todorov, 1999). As shown in Figure 1E, the ANOVA revealed significant effects of recency, similarity and their interaction (recency: $F(4, 576.98) = 59.20$, $p < .001$; similarity: $F(1, 382.40) = 28.18$, $p < .001$; interaction:

⁴ A new item could be counted twice if it lies between two old items that belong to different study-test lag bins.

⁵ The calculation of A_z is as follows. Consider r possible ratings, which yields $r - 1$ different decision thresholds. This process provides $r - 1$ pairs of hit rates and false alarm rates. Then, we compute the z -scores of those hit rates and false alarm rates and fit a line in z -space, known as the z -ROC curve. Finally, we define A_z using the intercept and the slope of the z -ROC curve:

$$A_z = \Phi \left[\frac{\text{intercept}}{\sqrt{1 + \text{slope}^2}} \right].$$

$F(4, 723.96) = 4.95, p < .001$). The Linear Mixed Model showed that a high similarity level leads to significantly lower A_z than a low similarity level for Bin 2 ($t(1608.26) = -2.36, p = .018$), Bin 3 ($t(1524.18) = -3.66, p < .001$), and Bin 4 ($t(1014.86) = -3.43, p < .001$), but not for Bin 1 ($t(1508.34) = -1.17, p = .244$) and Bin 5 ($t(236.06) = -0.19, p = .850$).

Simulation 1

To simulate data from Experiment 1, we had CMR-IA perform a continuous recognition test on the actual sequences of items used in the experiment. For each item, CMR-IA compared the current context with the context retrieved by the item, following Equation 4. If the context similarity exceeds the item recognition threshold c_{thresh}^{item} , the model would respond “old”; otherwise, it would respond “new”. To allow dynamic adjustment of the recognition threshold based on the expected distribution of similarities, we set c_{thresh}^{item} to be an exponentially weighted average of previous context similarities, scaled by the parameter c_{recog}^{item} (see Appendix A for more details). The model then encoded this item by updating the context and the M^{FC} matrix following Equation 2 ($\beta = \beta_{enc}$) and Equation 1. The context also drifted between each trial, following Equation 2 ($\beta = \beta_{post}^{recall}$). We simulated multiple sessions, each representing one of the subjects included in our analysis, and we used particle swarm optimization to identify the parameter values that minimized the chi-squared loss (sum of squared deviations normalized by the variance of each observed data point; Table 1). We calculated the model-predicted A_z by applying varying thresholds on context similarity, analogous to confidence judgments in the actual experiment.

As shown in Figure 1, CMR-IA demonstrates both recency and similarity effects, mirroring findings from the experiment. The model also predicts an interaction between recency and similarity, where a high similarity level decreases A_z specifically at large study-test lags. The contextual drift mechanism in CMR-IA accounts for the recency effect, as increased study-test lag diminishes context similarity. For the similarity effect, preceding same-category items enhance the representation of the current item in the

context, leading to increased context similarity and subsequently raising both hit rates and false alarm rates. However, for an old item, intervening same-category items also accelerate contextual drift. According to Equation 2, same-category items reduce the value of ρ , increasing the rate of contextual drift, resulting in a more rapid decline in recognition performance in the presence of similar intervening items⁶.

Simulation 2: Contiguity Effects

G. Schwartz et al. (2005) asked whether temporally defined associations play a role in item recognition. Using travel photographs as stimuli, they found that subjects were more likely to recognize a test item as old when the preceding old test item was studied in a temporally proximate position. In their experiment, subjects ($N = 91$) each studied six lists of 64 photographs and then completed an item recognition test on the same 64 studied photographs intermixed with 64 new photos drawn from the same pool. They compared recognition responses to old photos following those from adjacent list positions ($|\text{lag}| = 1$) versus those following photos from remote positions ($|\text{lag}| > 10$). Adjacent successive probes showed superior recognition performance relative to those from remote positions (Figure 2A and Figure 2C).

In our simulation, we replicated this experimental paradigm to assess whether CMR-IA exhibits a successive-probe contiguity effect. Each stimulus was represented as an orthonormal basis vector. To capture potential semantic associations between photos, we initialized M^{FC} as a random symmetric matrix with elements uniformly distributed between -0.1 and 0.1 and diagonal elements fixed at 0. The simulated subject studied 64 items in sequence, encoding each item as in Simulation 1. Between the study and test phases, context drifted following Equation 2 ($\beta = \beta_{\text{distract}}$). In the test phase, we presented 128 items sequentially, with half from the study phase. The simulated subject performed item recognition following Equation 4. We treated context similarity as the confidence

⁶ Same-category items tend to have a higher similarity with the current context $\mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}}$. Let $x = \mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}}$. Given that $0 < \beta < 1$, $\frac{d\rho}{dx} = -\beta + \frac{\beta^2 x}{\sqrt{1-\beta^2+\beta^2 x^2}} < 0$.

rating⁷. After each recognition judgment, the model updated context using the context input retrieved by the probe item, following Equation 2 ($\beta = \beta_{cue}$). Context also drifted between lists following Equation 2 ($\beta = \beta_{post}^{recall}$). We conducted 91 sessions with CMR-IA, employing the actual stimulus sequences from the experiment, and we used particle swarm optimization to identify the parameter values that minimized the sum of squared deviations (Table 1).

To evaluate the effect of temporal contiguity on recognition performance, we replicated the analysis conducted by G. Schwartz et al. (2005). Figure 2B and Figure 2D present the data generated by CMR-IA⁸. The model demonstrates a successive-probe contiguity effect, capturing the findings of G. Schwartz et al. (2005). The effect arises from the contextual update following the recognition judgment in CMR-IA, governed by β_{cue} . This mechanism integrates the context retrieved by the test item into the current context, enhancing its similarity with the contexts retrieved by adjacently studied items, thereby facilitating their recognition. Parameter β_{enc} also influences this effect, as it determines the degree of similarity among contexts associated with adjacent items. Increasing either β_{cue} or β_{enc} enhances the contiguity effect.

Osth and Fox (2019) investigated whether an analogous contiguity effect appears in associative recognition. In their experiment, subjects ($N = 112$) studied 60 word pairs and subsequently completed an associative recognition test that included 30 intact pairs and 30 rearranged pairs. The experimenters constructed the rearranged pairs such that one to five pairs separated their constituent items during the study phase. The findings revealed that false alarm rates (FARs) remained statistically equivalent across all lags, indicating no evidence of a contiguity effect in associative recognition (Figure 3A and Figure 3C).

We simulated this associative recognition paradigm using CMR-IA. Each simulated

⁷ We did not introduce c_{thresh}^{item} here, because we do not need a binary response for ROC analysis in this simulation and CMR-IA updates the context in the same way regardless of the judgment.

⁸ We constructed a “local” false alarm rate separately for adjacent and remote lags by incorporating data from new items contiguous with adjacent or remote old-item pairs, consistent with the methodology of G. Schwartz et al. (2005).

subject studied 60 sequentially-presented pairs, encoding each pair as in the preceding simulation. Then, after a distractor-filled delay ($\beta = \beta_{distract}$), we presented 60 test pairs (30 intact and 30 rearranged) to the simulated subject. The simulated subject performed associative recognition by updating the current context using the first item ($\beta = \beta_{cue}$) and then comparing the current context with the context input retrieved by the second item following Equation 5. If the context similarity exceeded the associative recognition threshold c_{thresh}^{assoc} , the simulated subject would respond “intact”; otherwise, it would respond “rearranged”. After the recognition judgment, the model updated the context with the second item ($\beta = \beta_{cue}$). Context also drifted between each list ($\beta = \beta_{post}^{recall}$). We simulated 111 sessions using the actual stimulus sequences from the experiment (omitting one non-completing subject). We employed particle swarm optimization to identify parameters that minimized the chi-squared loss (Table 1).

To clarify the predicted relationship between FARs and lag, we introduce the following simplifying assumptions. CMR-IA represents items as orthonormal basis vectors and represents pre-experimental associations via the weights in the M^{FC} matrix. Here we set these weights to zero. Consider study pairs $A_i - B_i, A_{i+1} - B_{i+1}, \dots, A_{i+L} - B_{i+L}$, where L denotes lag. According to Equation 2, while studying $A_i - B_i$, the context $\mathbf{c}_i = \rho \mathbf{c}_{i-1} + \frac{1}{\sqrt{2}} \beta_{enc} (\mathbf{f}_{A_i} + \mathbf{f}_{B_i})$; while studying $A_{i+L} - B_{i+L}$, the context $\mathbf{c}_{i+L} = \rho^L \mathbf{c}_i + \frac{1}{\sqrt{2}} \beta_{enc} \sum_{l=1}^L \rho^{L-l} (\mathbf{f}_{A_{i+l}} + \mathbf{f}_{B_{i+l}})$, where $\rho = \sqrt{1 - \beta_{enc}^2}$. During the test phase, we probe the model with the rearranged pair $A_i - B_{i+L}$. The model performs associative recognition by calculating the context similarity $r^{assoc} = \rho_t \mathbf{c}_t \cdot \mathbf{c}_{B_{i+L}}^{\text{IN}} + \beta_{cue} \mathbf{c}_{A_i}^{\text{IN}} \cdot \mathbf{c}_{B_{i+L}}^{\text{IN}}$ (Equation 5). For simplicity, we ignore the first term, which represents recency, and focus on the second term, which compares the context inputs of A_i and B_{i+L} . Because item representations are orthonormal, the dot product of distinct items evaluates to zero ($\mathbf{f}_X \cdot \mathbf{f}_Y = 0$ for $X \neq Y$). Furthermore, because the context $\mathbf{c}_i$ contains no features of the

future item B_{i+L} , their dot product is also zero ($\mathbf{c}_i \cdot \mathbf{f}_{B_{i+L}} = 0$). This yields:

$$\begin{aligned}
\mathbf{c}_{A_i}^{\text{IN}} &\propto M^{FC} \mathbf{f}_{A_i} = (1 - \gamma_{FC}) \mathbf{f}_{A_i} + \gamma_{FC} \mathbf{c}_i \\
\mathbf{c}_{B_{i+L}}^{\text{IN}} &\propto M^{FC} \mathbf{f}_{B_{i+L}} = (1 - \gamma_{FC}) \mathbf{f}_{B_{i+L}} + \gamma_{FC} \mathbf{c}_{i+L} \\
r^{\text{assoc}} &\propto \beta_{cue} \left((1 - \gamma_{FC}) \gamma_{FC} \mathbf{f}_{A_i} \cdot \mathbf{c}_{i+L} + \gamma_{FC}^2 \mathbf{c}_i \cdot \mathbf{c}_{i+L} \right) \\
&= \beta_{cue} \left((1 - \gamma_{FC}) \gamma_{FC} \rho^L \frac{1}{\sqrt{2}} \beta_{enc} + \gamma_{FC}^2 \rho^L \right) \\
&= \beta_{cue} \gamma_{FC} \rho^L \left(\frac{1}{\sqrt{2}} \beta_{enc} (1 - \gamma_{FC}) + \gamma_{FC} \right).
\end{aligned}$$

Because $\rho = \sqrt{1 - \beta_{enc}^2} < 1$ and β_{cue} , γ_{FC} , and β_{enc} are constrained between 0 and 1, an increase in lag L should theoretically diminish context similarity and thereby reduce FAR. Nonetheless, the simulation demonstrates that CMR-IA can yield comparable FARs for rearranged pairs at all lags (Figure 3B and Figure 3D) under certain parameter settings ($\beta_{enc} = 0.165$, $\gamma_{FC} = 0.046$, and $\beta_{cue} = 0.504$). This outcome occurs because low values of β_{enc} and γ_{FC} render the impact of L negligible, overshadowed by other sources of variance such as semantic associations and recency. When one increases these parameters (especially β_{enc}), CMR-IA produces a substantial contiguity effect. Thus, the absence of a contiguity effect in associative recognition does not refute the model (Hong, Gunn, Fazio, & Polyn, 2024).

Simulation 3: Differential Forgetting of Items and Associations

Hockley (1992) found that subjects forget item information more rapidly than associative information. In his study, subjects performed a variation of the continuous recognition task. Each list comprised 160 study-pair presentations, 80 single-item recognition tests, and 80 associative recognition tests intermixed in a single presentation stream. Subjects made intact-rearranged judgments on pairs and old-new judgments on single words. The experimenter constructed rearranged pairs by combining the left (or right) member of a study pair with the right (or left) member of an immediately preceding

study pair. Test probes appeared at study-test lags of 2, 4, 6, 8, and 16 intervening presentations. Hockley (1992) demonstrated that d' for item recognition declines more sharply than for associative recognition (Figure 4A and Figure 4C).

We simulated this paradigm with minor modifications. Whereas each subject in the original experiment completed 10 lists (two on each of five daily sessions), we simulated a single list in which study and test events strictly alternate: on each of 160 trials, the model first encodes a word pair and then receives a test probe. Instead of the Toronto Word Pool used by Hockley (1992), we used the PEERS Word Pool (Kahana, Lohnas, et al., 2024). The model encoded each pair as in the preceding simulation. For each test probe, the model performed item recognition if the test probe was a single word and associative recognition if it was a pair, following the procedures described in earlier simulations. After each judgment, the model updated the context with the test item or the second item in the test pair ($\beta = \beta_{cue}$). Context also drifted between trials ($\beta = \beta_{post}^{recall}$). We conducted a total of 300 simulation sessions and used particle swarm optimization to identify the parameter values that minimize the sum of squared deviations (Table 1).

CMR-IA exhibits a similar pattern of results to that reported in Hockley (1992). Both the experimental data and model simulations demonstrate differential forgetting rates, with d' declining more sharply for item than for associative recognition (Figure 4). In CMR-IA, differential forgetting rates reflect the differing mechanisms of item recognition and associative recognition. In item recognition, the model directly compares the retrieved context $\mathbf{c}_i^{\text{IN}}$ with the current context $\mathbf{c}_t$ (Equation 4), where $\mathbf{c}_t$ is substantially affected by contextual drift over time (controlled by parameters β_{enc} , β_{cue} , and β_{post}^{recall}). Conversely, in associative recognition (Equation 5), the model first reinstates the context retrieved by one item $\mathbf{c}_{i_1}^{\text{IN}}$ into the current context (controlled by β_{cue}). The model then compares the updated context $\mathbf{c}'_t = \rho_t \mathbf{c}_t + \beta_{cue} \mathbf{c}_{i_1}^{\text{IN}}$ with the context retrieved by the other item $\mathbf{c}_{i_2}^{\text{IN}}$. This comparison yields two terms: $\rho_t \mathbf{c}_t \cdot \mathbf{c}_{i_2}^{\text{IN}}$ and $\beta_{cue} \mathbf{c}_{i_1}^{\text{IN}} \cdot \mathbf{c}_{i_2}^{\text{IN}}$. The first term resembles item recognition, influenced by contextual drift but scaled by $\rho_t < 1$; the second term remains

unaffected by contextual drift, as $\mathbf{c}_{i_1}^{\text{IN}}$ and $\mathbf{c}_{i_2}^{\text{IN}}$ represent contexts stored at encoding and retrieved via M^{FC} . Thus, CMR-IA predicts more rapid forgetting for item than for associative information.

Simulation 4: Word Frequency Effects in Recognition Memory

In studies of item recognition, subjects more readily recognize rare words as targets and more easily reject them as lures, compared with common words Lohnas and Kahana (2013) (Figure 6A). Following Monaco, Abbott, and Kahana (2007), we hypothesized that the distinct semantic associations of rare words compared to common words contribute to the word frequency effect. Indeed, our analysis revealed a negative correlation between a word’s average semantic associations with other words, $\mathbf{s}^{\text{mean}}$, and its log word frequency ($p < .001$, $R_{adj}^2 = 0.058$; Figure 5). This finding aligns with Monaco et al’s finding that rare words possess more close neighbors in semantic space than common words.

We tested this hypothesis in a simulation of item recognition using standard experimental procedures. In each simulation, the model studied a random list of 100 words, followed by a test phase with 200 words – half old and half new (Schulman, 1967). We randomly selected these words from the 984 words used by Lohnas and Kahana (2013). We modeled encoding and retrieval as in earlier simulations of item recognition. Following Lohnas and Kahana (2013), we binned words into ten frequency bins and calculated hit rates and false alarm rates for each bin. We conducted 1000 simulation sessions and used particle swarm optimization to identify the parameter values that minimized the chi-squared loss (Table 1).

This simulation revealed that CMR-IA does not produce a word frequency effect. Given that common words typically exhibit weaker semantic associations with other words, the similarity between the context retrieved by the probe item and the current context is lower, so CMR-IA predicts a decline in both hit rate (HR) and false alarm rate (FAR) as the word frequency increases (Figure 6B). Furthermore, we did not observe superior recognition performance for rare words compared to common words in CMR-IA. In this

simulation, performance did not vary meaningfully between the lowest word frequency bin ($d' = 1.89$) and the highest word frequency bin ($d' = 2.18$). This null result aligns with prior theoretical work showing that the word frequency effect depends on some differential processing of rare and common words (e.g., Glanzer, Hilford, & Kim, 2004; Shiffrin & Steyvers, 1997). In particular, a large empirical literature has suggested that rare words attract greater attention than common words, resulting in stronger stored associations (Malmberg & Nelson, 2003; Rao & Proctor, 1984). Researchers have also argued that subjects make trial-by-trial adjustments to their recognition criteria, increasing their threshold for rare words (Murdock, 1998; Benjamin, Diaz, & Wee, 2009; Monaco et al., 2007).

We incorporated both mechanisms into CMR-IA using $\mathbf{s}^{\text{mean}}$ as an indicator of word frequency⁹. When updating the M^{FC} matrix, we added an attention mechanism by adding an attention coefficient ψ into Equation 1, enhancing the association of rare words with the context. When making old-new judgments, we adjusted the recognition threshold $c_{\text{thresh}}^{\text{item}}$ to be sensitive to the frequency of the test word according to Equation A10 (see Appendix A for more details). We refit CMR-IA with these additional mechanisms (1000 sessions; Table 1).

With these additional mechanisms, CMR-IA predicts an increase in FAR and a decrease in HR as word frequency rises (Figure 6C), replicating the word frequency effect. The parameter c_s is pivotal for reducing the FAR for rare words, as it determines how sensitive the recognition threshold is to word frequency, thereby controlling the slope of FAR. However, a positive c_s , while decreasing FAR for rare words, simultaneously reduces their HR. To achieve a higher HR for rare words, the attention coefficient ψ , determined by the parameters ψ_s and ψ_c , becomes essential. Specifically, ψ_s governs the sensitivity of the attention coefficient to word frequency, thus controlling the slope of HR¹⁰.

⁹ This approach avoids introducing extraneous input information compared to using an explicit representation of word frequency.

¹⁰ As detailed in Appendix A, the proposed attentional boost for rare words only affects the

Monaco et al. (2007) produced the word frequency effect using a Hopfield network with only a threshold adjustment mechanism, without the attention mechanism. They posited that rare words are more proximal to strong attractors due to differences in semantic associations, resulting in a higher recognition accuracy and rendering the attention mechanism unnecessary. In contrast, CMR-IA does not yield higher accuracy for rare words without the attention mechanism (Figure 6B). We conducted additional control simulations to test whether either mechanism alone could produce the mirror effect. When incorporating only the threshold adjustment mechanism without the attention mechanism, CMR-IA predicted a mean slope of HR at 0.004 and a mean slope of FAR at 0.012. Conversely, incorporating only the attention mechanism without the threshold adjustment mechanism yielded a mean slope of HR at -0.014 and a mean slope of FAR at -0.004 . Neither attempt succeeded in producing a mirror effect. Thus, both the threshold adjustment mechanism and the attention mechanism appear necessary for CMR-IA to capture the word frequency effect.

Simulations 5-8: Recency, Similarity, and Contiguity in Cued Recall

As our work aimed to develop a CMR-based model of both item and associative information, we now turn to the analysis of data from cued recall of paired associates, as this paradigm depends explicitly on the retrieval of associations between the cue and the target. Simulation 5 investigates serial position effects in memory for paired associates, utilizing a classic dataset from Murdock (1963). Simulation 6 examines CMR-IA's prediction regarding data on associative symmetry, fitting data described in Kahana (1993) and Kahana (2002). Simulation 7 explores data on prior-list and intra-list intrusions in cued recall (Davis, Geller, Rizzuto, & Kahana, 2008), focusing on the effects of recency and contiguity. Simulation 8 looks at similarity effects and their role in producing intra-list

item-to-context associative matrix (M^{FC}); it does not alter the context-to-item associative matrix (M^{CF}) that drives the retrieval process in free recall. Thus, our model does not predict a low-frequency recall advantage. We note that the empirical picture is complex, as Lohnas and Kahana (2013), for example, demonstrated a recall advantage for both low-frequency and high-frequency words.

intrusions, utilizing data from a study of memory for name-face pairs (Pantelis, van Vugt, Sekuler, Wilson, & Kahana, 2008). In a subsequent section, we evaluate CMR-IA’s ability to account for the interactions between item and associative information, as seen in data from successive testing procedures (Kahana, 2000).

Simulation 5: Serial Position Effects in Cued Recall

In studies of memory for associations, researchers often present subjects with a short series of paired items (typically words) and then test each of the studied pairs in a random order (e.g., Tulving & Arbuckle, 1963). These experiments typically exhibit very modest serial position effects. However, a pronounced recency effect appears when considering the first tested pair. For example, Figure 7A illustrates a significant recency effect in a paired-associate experiment where subjects ($N = 16$) attempted to recall a single probed pair after studying a six-pair list (Murdock, 1963).

Here we investigate whether CMR-IA can replicate the recency effect observed in Murdock’s (1963) experiment. We utilized the PEERS Word Pool (Kahana, Lohnas, et al., 2024) to simulate Murdock’s experimental procedure with CMR-IA. Each simulation session included 78 lists, mirroring the original experiment’s design. During the study phase, the simulated subject studied six word pairs in sequence, encoding each pair by updating the context along with the M^{FC} and M^{CF} matrices following Equation 2 ($\beta = \beta_{enc}$) and Equation 1. Between the study phase and test phases, context drifted following Equation 2 ($\beta = \beta_{distract}$). In the test phase, we randomly selected a studied pair and presented the first item as a cue. The simulated subject performed cued recall by first updating the current context with the cue and then attempting to retrieve an item from memory, following Equation 6 ($\beta = \beta_{cue}$). Following successful recall, the simulated subject updated the context with the retrieved item following Equation 2 ($\beta = \beta_{rec}$). Contextual drift also occurred between each list following Equation 2 ($\beta = \beta_{post}^{recall}$). We conducted a total of 100 sessions and employed particle swarm optimization to identify parameter values that minimized the sum of squared deviations (Table 1).

We replicated the recency effect in cued recall using CMR-IA (Figure 7B). The contextual drift during the study-test lag, governed by β_{enc} and $\beta_{distract}$, determines this effect. The context associated with more recently studied pairs overlaps more with the current context, leading to improved recall performance.

Simulation 6: Associative Symmetry

One of the classic questions concerning associations relates to their directionality. According to Ebbinghaus (1885/1913), studying a sequence of items leads to stronger forward than backward associations. Indeed, the concept of directional associations goes back to Herbart’s (Herbart, 1834) theory of associative learning. According to this view, representations of the first and the second members of a pair associate via a directional bond that is stronger in the forward direction than in the backward direction. An opposing view, however, emerged in Gestalt theories of associations. Kohler (1940) summarized the Gestalt perspective, which saw associative learning as creating a new memory composed of the elements of each to-be-associated item. This new memory does not possess any forward directionality but rather combines the two constituents into a new “whole”. Asch and Ebenholtz (1962) argued that empirical findings on associative recall align more closely with this Gestalt hypothesis, citing early work on backward associations by Murdock (1956). For a summary of the empirical literature on associative symmetry, see Kahana (2002).

To investigate the directionality of associative recall in CMR-IA, we modified Simulation 5 by probing memory in either the forward or backward direction. Each studied pair received a preassigned testing direction: presenting item A_i as a cue to recall item B_i for forward direction, or presenting item B_i as a cue to recall item A_i for backward direction. We tested these pairs in the reverse order of their initial presentation¹¹.

Figure 8B illustrates CMR-IA’s predicted probabilities for forward versus backward recall.

¹¹ To be comparable to the experiment from Kahana (1993), each simulation session consists of 42 lists. Other details remain the same as Simulation 5.

For comparison, Figure 8A presents empirical data reported by Kahana (1993). CMR-IA predicts equivalent forward and backward recall, consistent with experimental data. The model predicts symmetry despite predicting forward asymmetry in free recall, as both items in a pair become associated with a single representation of context. In contrast, when simulating lists of discretely presented items, CMR updates the context after each encoding event.

Kahana (2002) suggested a stronger test of the associative symmetry hypothesis. Specifically, if subjects encode forward and backward associations holistically, then successively testing these associations should exhibit similar correlations when the repeated tests involve congruent cues (e.g., cueing with A for recall of B , or vice versa, on both tests) compared to incongruent cues (cueing with A for B on the first test, and B for A on the second test; or vice versa). Using this successive testing procedure, Kahana (2002) evaluated the correlation between successful retrieval of a given association when probed in the same direction (congruent cues) or in different directions (incongruent cues). This is one of several applications of the method of successive tests (see (Kahana, 2000), and also Section 4).

Here, we examined CMR-IA’s predictions under these conditions. Each simulation session consisted of six lists. During the study phase, we presented 12 word pairs, randomly sampled from the PEERS Word Pool (Kahana, Lohnas, et al., 2024). The model encoded each pair and performed cued recall as in Simulation 5. Then, we tested all the studied pairs twice across two test phases, referred to as Test 1 and Test 2. In Test 1, we cued half of the studied pairs in the forward direction ($A - ?$), and the other half in the backward direction ($? - B$). In Test 2, we maintained half of the pairs in the same direction as Test 1 while reversing the direction for the remaining pairs. We defined *congruent* conditions as those in which both tests are same-direction (forward-forward and backward-backward) and *incongruent* conditions as those involving different directions (forward-backward and backward-forward). Additionally, context drifted between phases

following Equation 2 ($\beta = \beta_{distract}$) and between lists ($\beta = \beta_{recall}^{post}$).

As this simulation involved repeated tests, we incorporated output encoding as described at the end of Section 2. Specifically, upon successful recall, the model updates M^{FC} and M^{CF} following Equation 1. We use output encoding in all simulations that involve repeated test events.

We conducted the simulation over 100 sessions and used particle swarm optimization to identify the parameter values that minimized the sum of squared deviations (Table 1). As shown in Table 2, CMR-IA predicts similar correlations for congruent and incongruent cues, consistent with the findings of Kahana (2002).

Simulation 7: Prior-list and Intra-list Intrusions in Cued Recall

Intrusions in cued recall tend to come from the target list more often than from prior lists (we refer to these two types of intrusions as intra-list intrusions, ILIs, and prior-list intrusions, PLIs). PLIs exhibit a recency effect insofar as they tend to come from recent lists more often than from remote lists (Davis et al., 2008). ILIs exhibit a contiguity effect insofar as they tend to come from positions near that of the just tested pair, an effect also observed in serial recall (Solway, Murdock, & Kahana, 2012) and probed recall tasks (Kahana & Caplan, 2002). Figure 9A and Figure 9C illustrate the conditional-response probability for ILIs and the recency effect for PLIs in Davis et al.’s (2008) experiment.

Here, we investigate whether CMR-IA exhibits these basic patterns. We simulated the experimental paradigm of Davis et al. (2008). In each simulation session, we presented 16 lists, each containing 12 word pairs sampled from the PEERS Word Pool (Kahana, Lohnas, et al., 2024). Following the presentation of each list, we tested the simulated subject on eight of the studied pairs. Encoding, cued recall, and context updating followed the same procedures as in earlier simulations. We randomly probed half of the tested pairs in the forward direction and the remainder in the backward direction. We conducted the simulation 500 times and used particle swarm optimization to identify parameter values that minimize the chi-squared loss (Table 1).

CMR-IA effectively reproduces the PLI recency effect (Figure 9B) and the ILI contiguity effect (Figure 9D), while also matching the cued recall accuracy and the overall rates of PLIs and ILIs observed in the data. The rate of contextual drift, governed by various β parameters, determines the extent to which PLIs from more recent lists prevail in retrieval competitions compared to those from more distant lists. A word pair from a more recent list associates with a context more similar to the current context, thereby enhancing the likelihood of retrieval. For ILIs, the rate of contextual drift determines the similarity among the contexts associated with neighboring studied pairs, with a higher rate of contextual drift leading to a more pronounced ILI contiguity effect.

Simulation 8: Similarity Effects

Simulation 8 examines the effects of similarity on ILIs. Whereas most prior studies using verbal materials manipulated similarity in a categorical manner (Osgood, 1949), we sought to test CMR-IA using a dataset that varied similarity parametrically. We simulated data from a name-face association memory study where researchers synthesized faces using a perceptual model that establishes a metric space for calculating inter-item similarity (Pantelis et al., 2008). In their experiment, subjects ($N = 25$) learned a set of eight name-face pairs across ten repeated study-test trials. They then examined both the probability of a correct recall and the probability of ILIs as a function of facial similarity.

We simulated the experimental paradigm to evaluate how similarity influences cued recall in CMR-IA. The model learned a list of eight randomly chosen face-name pairs drawn from a pool of 16 faces and 16 names, as outlined in Pantelis et al. (2008). We employed basis vectors to represent names and faces and simulated inter-item similarity through the pre-experimental associations in M_{pre}^{FC} and M_{pre}^{CF} . We calculated cosine similarity between faces using the coordinates provided in the multidimensional scaling (MDS) space from Pantelis et al. (2008). We assumed no pre-existing associations among names or between names and faces. We did not simulate the multi-trial learning effects reported by Pantelis et al. (2008), focusing instead on the encoding and retrieval of a once-presented list. The

model encoded each studied pair as in prior simulations, and context drifted between study and test phases ($\beta = \beta_{distract}$). In the test phase, we probed the model with each studied face to determine which name it would recall (if any). The model performed cued recall as in Simulation 5¹². We performed the simulation 10,000 times and used particle swarm optimization to identify parameter values that minimized the chi-squared loss (Table 1).

Pantelis et al. (2008) investigated whether cueing memory with a face possessing many neighbors (i.e., similar faces) among the studied name-face pairs leads to poorer memory for the associated name. They defined neighbors as faces that fall within a radius of three units of the probe face in the MDS space. They found that recall probability decreased monotonically with the number of neighboring faces in the list, a phenomenon referred to as the “neighborhood” effect (Figure 10A). Furthermore, they examined whether ILIs are more likely to originate from neighboring faces. By calculating the Euclidean distance between the cue face and the face associated with the incorrectly recalled name, they found that the conditional probability of making an ILI decreased monotonically with the distance between the two faces (Figure 10C). These results align with the predictions of CMR-IA (Figure 10B and Figure 10D). In CMR-IA, pre-experimental associations in M^{FC} , governed by the parameter s_{FC} , represent similarity among faces. During cued recall, the simulated subject first updated the current context with the test face, thereby strengthening representations for the test face’s neighbors. An increase in the number of neighbors heightens the likelihood of confusion during name recall. Furthermore, a neighbor’s proximity in the face space determines the strength of its representation in the updated context. Consequently, the simulated subject is more likely to mistakenly retrieve the paired name of a closer neighbor.

¹² We prohibited CMR-IA from recalling faces by allowing only names to enter the accumulator during this simulation. This also implies that s_{CF} , which governs the pre-experimental associations between faces in M^{CF} , has no mathematical influence in this simulation.

4. Successive Tests of Item and Associative Information

Here we consider the relation between item and associative information as measured using the successive testing technique (Kahana, 2000). This method enables researchers to measure the correlation between successful retrieval of a given item under different conditions. In Tulving’s classic procedure (Flexser & Tulving, 1978), subjects studied a list of $A - B$ word pairs and were then tested successively, first by item recognition and then by cued recall. In the item-recognition test, subjects were shown list items (B items from each of the studied pairs) intermixed with non-list items. Subjects responded “yes” to items if they remembered seeing them in the study list. In the cued-recall test, subjects attempted to recall the B items when given the A items as cues. In this manner, memory for each of the B items was tested twice — first by recognition and then later by recall. Flexser and Tulving observed a moderate degree of dependency between item recognition and cued recall at the level of individual subject-items. Expressed using Yule’s Q , a measure of association for 2×2 contingency tables, one finds values ranging from 0.45 to 0.65 across a wide range of experiments (Kahana et al., 2005)¹³.

Going beyond Tulving’s recognition-recall analysis, we examine contingencies between successive tests of item and associative information. Comparisons of recognition and recall can be simplified by separating differences due to the response stage from differences in the information being tested. This can be achieved by making two comparisons, first between item recognition and cued recall, and then between associative recognition and cued recall. Differences between item recognition and cued recall reflect both differences in information and differences in the response stage. Differences between associative recognition and cued recall should primarily reflect differences in the response stage. To the extent that the differences between item recognition and cued recall exceed those between associative recognition and cued recall, these can be attributed to differences

¹³ Yule’s Q behaves much like an ordinary Pearson correlation, ranging from -1.0 (perfect negative correlation) to +1.0 (perfect positive correlation). In this paper, Q is determined separately for each subject, with 0.5 added to each of the cells in the contingency table prior to calculating the Q value.

in the type of information tested in the two tasks.

Experiment 2

Experiment 2 compared contingency relations between three types of recognition tasks and a subsequent cued recall test for the same items: In the associative recognition task, subjects judged probe pairs as intact or rearranged; in the item recognition task, subjects judged individual items as old or new; in the pair recognition task, subjects judged pairs of items as old or new. Although our central question concerned the difference in correlation between recognition and recall under these three conditions, we also examined the hypothesis that asymmetrical associations may underlie the recognition-failure of recallable words (Rabinowitz, Mandler, & Barsalou, 1977). According to this account, the backward association from B to A is significantly weaker than the forward association, and therefore, when tested on recognition of B alone, subjects often fail to recall A , resulting in an incomplete match to the stored information. As expressed by Rabinowitz et al. (1977, p. 661) “Retrieval must be asymmetric in order for recognition failure to occur, and it is expected to be asymmetric given the nature of the instructions (study the pair so that you can recall the TBR [to-be-remembered item] given the LC [context item] as a cue).” We did not expect to find support for this hypothesis because of the extensive body of literature showing symmetric recall of paired associates under a wide range of experimental conditions (see Kahana, 2002, for a review). Nonetheless, none of these studies had specifically considered symmetry in the recognition-failure paradigm, so it seemed worthwhile to explore.

We tested the asymmetry hypothesis by manipulating the order of the pairs in each condition of this experiment. In the cued recall phase, subjects were either shown A and asked to recall B , or they were shown B and asked to recall A . Order effects were also tested in the three recognition tests. In pair and associative recognition, subjects were either shown $A - B$ or $B - A$, and in item recognition, subjects were either tested on the A item or the B item. It was hypothesized that symmetry would be obtained in cued recall

and item recognition but not in associative or pair recognition. The hypothesized violations of symmetry in the pair and associative recognition tests were motivated by the belief that all of the information about the stimulus (including perceptual features) is encoded at study and enters the memory comparison at test. When the full pair is shown at test and it is reversed in order from the presentation at study, the mismatching perceptual information should lower performance on the recognition test. This should not happen in recall or item recognition because only a single item is provided as the test cue.

Methods

Subjects

Ninety undergraduate students at the University of Toronto participated for extra credit in an introductory psychology course. All subjects were native English speakers.

Procedure

We randomly assigned subjects to each of three groups. Group one performed a standard yes-no item recognition test. Group two performed the pair recognition test used by Wallace (1978). Group three performed an associative recognition test. Twenty-one subjects recalled fewer than 10% of the list items and were excluded from the data analyses.

Subjects studied a list of 48 word-pairs and then took part in successive recognition and cued-recall tests. During the study phase, word pairs, randomly selected from the Toronto word pool, were presented visually at a rate of one pair every two seconds. We did not test subjects on the last eight word pairs on the study list, designating these as recency buffers.

During the test phase, subjects in group one (item recognition) made yes/no judgments on old and new single items. Subjects in group two (pair recognition) made yes/no judgments on intact and new pairs. Subjects in group three (associative recognition) made yes/no judgments on intact (old) and rearranged (new) word pairs. Intact pairs matched those shown during the study, whereas rearranged pairs combined items from two different study pairs to form a new pair. Half of the test probes were tested

in the forward order ($A - B$) and half in the backward order ($B - A$). If a subject did not respond within eight seconds, the computer moved along to the next item.

After completing the recognition-memory test, we administered a cued-recall test on the studied pairs. In the associative recognition group, we only tested intact pairs during the cued-recall phase. Half of the items were tested with forward recall; the other half were tested with backward recall. Subjects had 12 seconds to type their responses on a computer keyboard. The entire study-test process was repeated multiple times. On each repetition, a new random sample of words was selected from the Toronto word pool.

Results

Before turning to our primary questions concerning the dependency relations between successive recognition and recall tests, we comment briefly on the differences in overall recognition and recall performance (Figure 11A and Figure 11B). Performance on cued recall differed across the test groups ($F(2, 66) = 11.0, p < .001$). Tukey HSD post-hoc comparisons revealed significant differences between associative and pair recognition groups ($p < .05$), and between item and associative recognition groups ($p < .01$). The difference between item and pair recognition groups just missed our threshold for statistical significance ($p = .06$). These differences likely reflected the differing numbers of items intervening between the original study and cued recall for each of the three groups. For the associative recognition group, 40 pairs were tested in the recognition phase, for the pair recognition group, 80 pairs were tested in the recognition phase, and for the item recognition group, 80 individual items were tested in the recognition phase. Recognition performance, as measured by d' , also varied across the three groups (item recognition group: $d' = 1.56 \pm 0.07$; pair recognition group: $d' = 2.20 \pm 0.11$; associative recognition group: $d' = 1.46 \pm 0.13$; $F(2, 66) = 13.1, p < .01$). d' in the pair-recognition group was significantly higher than either the associative recognition or the item recognition groups ($p < .01$). In contrast, no significant difference in d' emerged between the item and associative recognition groups.

Dependency Relations

Both the associative and pair recognition tests yielded higher recognition-recall dependencies than the item recognition test (Figure 11C). Q varied significantly as a function of the type of recognition test ($F(2, 66) = 10.6, p < .001$). Tukey HSD post-hoc comparisons revealed that recognition-recall dependencies were significantly higher when associative recognition ($p < .001$) or pair recognition ($p < .05$) were tested than when item recognition was tested. The difference between the associative and pair recognition exhibited a trend suggestive of a potential difference between these conditions ($p = .08$).

Correlations between Recognition and Recall Performance across Subjects

In addition to examining item-by-item dependencies, one can consider the inter-subject correlations between recognition and recall performance. The correlation between associative recognition and cued recall ($r^2 = 0.80$) was higher than the correlation between pair recognition and cued recall ($r^2 = 0.74$) and the correlation between item recognition and cued recall ($r^2 = 0.33$). The item-recognition versus cued-recall correlation accords well with classic data reported by Underwood, Boruch, and Malmi (1978). All correlations were statistically significant.

Associative Symmetry

Although experimental conditions were designed to promote sequential processing of the pairs¹⁴, we failed to observe any significant difference between forward and backward cued recall; the mean difference score between forward recall and backward recall was 0.007 (± 0.020 [95% CI], with forward recall = 0.298 and backward recall = 0.291). Unlike cued recall, we observed significantly better recognition of intact probes than of reversed probes in the associative recognition task ($t(37) = 4.9, p < .001$). If the magnitude of asymmetry found in associative recognition is taken as an estimate of the asymmetry that might be expected in cued recall (if the null hypothesis were false), then the present data has a

¹⁴ Subjects read the word pairs aloud from left to right as soon as they appeared. During the recall phase, the probe word appeared on the same side of the screen as it had appeared during the study phase.

power of 0.93 for rejecting the null hypothesis of symmetry in cued recall. As a general rule, successful cued recall of meaningful pairs does not depend on the order of study (Kahana, 2002).

Our findings demonstrate a clear dissociation between associative recognition and cued recall. This dissociation contrasts with the high dependency relations and inter-subject correlations between associative recognition and cued recall. Nonetheless, one can explain the asymmetry effect by assuming that subjects base their recognition judgments partially on the perceptual familiarity of the pair. If one reverses the order of the pair, perceptual familiarity can not be used in making the recognition judgment. Memory models can account for this effect if item vectors include source information on the perceptual attributes of an item (e.g., Kahana, 2002; Polyn et al., 2009).

Manipulating recall direction allowed us to test the hypothesis that failed backward recall produces recognition-failure of recallable words (e.g., Rabinowitz et al., 1977). The present data argue against this explanation. Because typical levels of recognition failure were coupled with near-perfect symmetry between forward and backward recall, we do not find any evidence for the Rabinowitz et al. (1977) hypothesis.

Experiment 3

To more fully evaluate the relations between item and associative information, we conducted another experiment that compared successive recognition tests where the first test is either item or associative information, the second test is either item or associative recognition, and the items tested in the first and second tests are either the same or different. By introducing conditions of total informational overlap and no informational overlap, we can evaluate whether CMR-IA predicts the broad range of inter-task dependencies observed in experimental data.

Kahana et al. (2005) showed that the observed correlation between recognition and recall not only reflects the inherent model-derived correlation, but also reflects the effects of variability during encoding and retrieval, and the effect of the first test on the second test.

Variability during encoding will influence both item-specific and relational information, thereby boosting the observed correlation. Output encoding can, under certain assumptions, increase the correlation because information stored at test may differentially affect recognized and non-recognized items, thereby increasing the probability of recall. On the other hand, variability in the criterion for successful recognition or recall can reduce the correlation between tasks. This is because the two tests of a given pair are widely separated in time, and the retrieval conditions therefore introduce variance that is unique to each test. To attenuate the effect of encoding variability in this study we presented each study pair six times. We included comparisons of both identical probes (on Tests 1 and 2) and unique probes to allow us to directly estimate the effects of variability during retrieval on the Test 1 – Test 2 correlation.

Method

Subjects

Sixty-three undergraduate students at Brandeis University participated for extra credit in an introductory psychology course. All subjects were native English speakers.

Procedure

A single trial consisted of a study phase followed by two successive recognition test phases (termed Test 1 and Test 2). Each study list was composed of 25 unique word pairs (denoted $A - B$) drawn at random from the noun subset of the Toronto noun pool (Friendly, Franklin, Hoffman, & Rubin, 1982). Across the study list, each $A - B$ pair appeared six times, with the constraint that at least one distinct pair intervened between repeated pairs. Each pair appeared individually in the center of a computer screen for 2 s. After studying a given list and before the successive recognition tests, subjects made same-different discrimination judgments on 15 pairs of patterns. This distractor task served to attenuate recency effects. Each pattern was constructed by randomly filling elements in a 4×4 square matrix.

Successive recognition tests followed the distractor task. The cues given in these

tests reflected seven different within-subject conditions. Each of the 25 word pairs in the study list and an additional five pairs that were not in the study list were assigned to a condition, which pre-determined the cues given in Test 1 and Test 2. The non-studied pairs serve as lures that were tested in both Test 1 and Test 2. Five target conditions represent the following cues at Test 1 and Test 2: Six “Same Item” cues (A or B at Test 1 and the same item at Test 2); six “Different Item” cues (A or B at Test 1 and the other item at Test 2); three “Intact Pair” cues ($A - B$ at Test 1 and $A - B$ at Test 2); nine “Item/Pair” cues (A or B at Test 1, and the pair $A - B$ at Test 2); six “Pair/Item” cues (the pair $A - B$ at Test 1 and A or B at Test 2)¹⁵. In addition to testing the 25 studied pairs and the five lures that appeared in both Test 1 and Test 2 (“Repeated Lure”), there were 30 additional lures presented at each test (“Non-repeated Lure”). Of these 30 additional lures, half were single items and half were pairs.

In the recognition test phases, we asked subjects “Was the word on the list?” or “Was the word pair on the list?” If the probe item is a word, then either A , B , or X (where X is a lure not from the study list) appeared in the center of the screen. If the probe item is a word pair, then either $A - B$ or $X - Y$ (where X and Y denote lures) are presented in the center of the screen. Subjects responded “yes” or “no” by pressing the left and right control keys. For half of the subjects, the left key indicated a “yes” response; for the other half, the right key indicated a “yes” response.

The pattern-matching distractor task began after each of the recognition tests. The entire experiment consisted of three trials (each trial having a study list followed by two successive recognition tests as described above). We designated the first trial as practice and excluded these data from subsequent analyses. The stimuli for each trial consisted of a different random sample taken from the Toronto noun pool, with no words repeated across trials.

¹⁵ The five non-studied pairs were randomly assigned to one of the target conditions along with the 25 studied pairs. The following analysis only included the non-studied pairs that were tested with “Same Item” cues and “Intact Pair” cues. The result is represented by the “Repeated Lure” condition in Figure 12.

Results

Figure 11A and Figure 11B illustrate overall accuracy across conditions for each of the successive memory tests. For the first test, the proportion of “yes” responses (hit rate) depends primarily on whether the cue is a single item or a pair. For single-item cues, performance hovers around 0.81, whereas for pair cues, performance hovers around 0.90. This difference likely reflects the benefits of having a richer cue to compare with memory.

For the second test, the hit rate for single items not present on the first test drops precipitously (from 0.82 to 0.68), whereas the hit rate for items present on the first test increases slightly (from 0.81 to 0.82). For the second test, the hit rate for pairs present on the first test increases slightly (from 0.90 to 0.92). This pattern strongly suggests an effect of output encoding.

Dependency Relations

The Q values varied widely across the different Test 1 – Test 2 conditions (Figure 11C). If associative (pair) recognition and cued recall are more alike than they are different (e.g., Nobel & Shiffrin, 2001), the “Item/Pair” condition should yield data comparable to the values obtained in Experiment 2. Consistent with this position, Q for item recognition followed by associative (pair) recognition was 0.64 in this experiment, only slightly higher than the values obtained in Experiment 2 for comparisons of item recognition and cued recall.

When the order of item and associative tests is reversed, with the associative test given first (the “Pair/Item” condition), the correlation does not change significantly ($Q = 0.59$). Testing the same information on Test 1 and Test 2 yielded very high correlations ($Q = 0.94$ for the “Intact Pair” condition, and $Q = 0.86$ for the “Same Item” condition), with values exceeding the correlation between associative recognition and cued recall.

Subjects exhibited a significant positive contingency between successive recognition judgments made on the first and second members of a studied pair, even though the two

tests are widely separated in time (the “Different Item” condition, $\mathcal{Q} = 0.26$). Such a finding could be explained in one of two ways: 1) the correlation is induced by shared variability, across pairs, in the quality of encoding both the A_i and the B_i items. 2) when A_i is presented as a cue in Test 1, subjects implicitly recall the $A_i - B_i$ pair, thereby strengthening the representation of B_i in memory (output encoding).

5. Modeling the Dependency between Items and Associations

Here, we evaluate the fit of CMR-IA to data from Experiments 2 and 3. For both experiments, we developed simulations that closely matched the procedures used with actual subjects.

Simulating Experiment 2

During the study phase, we presented a list of 48 word pairs randomly selected from the PEERS Word Pool (Kahana, Lohanas, et al., 2024). The simulated subject encoded each pair by updating the context along with the M^{FC} and M^{CF} matrices following Equation 2 ($\beta = \beta_{enc}$) and Equation 1. For item recognition, the simulated subject compared the current context with the context input brought by the item, following Equation 4. If the context similarity exceeds the item recognition threshold c_{thresh}^{item} , the simulated subject would respond with “old”; otherwise, the response would be “new”. In the case of associative recognition and pair recognition, the simulated subject updated the current context using the first item following Equation 2 ($\beta = \beta_{cue}$), and then compared the current context with the context input brought by the second item following Equation 5. If the context similarity exceeds the associative recognition threshold c_{thresh}^{assoc} , the simulated subject would respond with “intact”; otherwise, the response would be “new/rearranged”. After making the recognition judgment, the simulated subject updated the context with the test item or the second item in the test pair, following Equation 2 ($\beta = \beta_{cue}$). For items or pairs judged as old, the simulated subject updated the M^{FC} and M^{CF} matrices following Equation 1, to simulate output encoding. In the case of cued recall, the simulated subject first updated the current context with the cue and then

attempted to retrieve an item from memory, following Equation 6 ($\beta = \beta_{cue}$). Upon successful recall, the simulated subject updated the context with the retrieved item according to Equation 2 ($\beta = \beta_{rec}$). The simulated subject also encoded the cue and the retrieved item as a pair following Equation 1. Additionally, context drifted between phases following Equation 2 ($\beta = \beta_{distract}$) and between study-test-test trials ($\beta = \beta_{post}^{recall}$). We conducted 100 simulation sessions for each group using particle swarm optimization to identify parameter values that minimize the sum of squared deviations (all groups shared the same set of model parameters; Table 1).

Figure 11 illustrates simulation results alongside the experimental data. Performance on cued recall differs significantly across simulation groups ($F(2, 297) = 433.20, p < .001$). Tukey HSD post-hoc comparisons reveal significant differences between all three groups ($ps < .001$). Recognition performance, as measured by d' , also demonstrates notable differences across three groups ($F(2, 297) = 266.43, p < .001$), with Tukey HSD post-hoc comparisons revealing significant distinctions among all groups ($ps < .001$). The dependency relations, assessed through $\mathcal{Q}$, vary according to the types of recognition test ($F(2, 297) = 77.00, p < .001$). Tukey HSD post-hoc comparisons indicate significant differences between all three groups ($ps < .001$). Although we report statistics on the model, we recognize that in modeling group data rather than individual subjects (as is commonly done), it is not straightforward to relate the statistical findings between the model and the data.

For item recognition, CMR-IA directly compares the retrieved context with the current context. In contrast, for both associative (pair) recognition and cued recall, the cue item or the first item in the test pair reinstates its associated context. Subsequently, CMR-IA employs the updated context to recall or recognize the other item. This congruence in how the model uses the stored association accounts for the high dependency between associative (pair) recognition and cued recall. The differences between item recognition and associative (pair) recognition also interact with the process of output

encoding. Upon item recognition, CMR-IA only encodes the recognized probe. Although the context input brought by the recognized probe involves the other item of the studied pair, the strengthening of their association is minimal. In contrast, upon associative (pair) recognition, the model explicitly encodes the recognized pair, thereby strengthening the association between the two items. Nevertheless, we note that CMR-IA lacks a different mechanism for pair recognition compared to associative recognition.

We also examine the hypothesis of associative symmetry in CMR-IA. Consistent with our Gestalt approach in modeling the encoding of word pairs, we find no significant difference between forward and backward in cued recall ($p = .925$). However, our model fails to replicate the empirically observed asymmetry in associative recognition tests ($p = .829$). In CMR-IA, an intact pair and a reversed pair are indistinguishable for the simulated subject. A revision of the model that encodes the location of the words on the screen within the item vectors could address this limitation.

Simulating Experiment 3

We replicated the same experimental paradigm to simulate Experiment 3 in CMR-IA. During the study phase, we presented a list consisting of 25 unique word pairs randomly selected from the PEERS Word Pool (Kahana, Lohnas, et al., 2024). Each pair appeared six times in a random order. The simulated subject encoded each pair by updating the context alongside the M^{FC} and M^{CF} matrices following Equation 2 ($\beta = \beta_{enc}$) and Equation 1. In the test phase, the simulated subject completed two successive recognition tests. For item recognition, the simulated subject compared the current context with the context input brought by the item, following Equation 4. If the context similarity exceeds item recognition threshold c_{thresh}^{item} , the simulated subject would respond with “old”; otherwise, the response would be “new”. For associative recognition, the simulated subject updated the current context using the first item following Equation 2 ($\beta = \beta_{cue}$) and then compared the current context with the context input brought by the second item following Equation 5. If the context similarity exceeds the associative

recognition threshold c_{thresh}^{assoc} , the simulated subject would respond with “old”; otherwise, the response would be “new”. We introduced a new parameter σ to model retrieval variability in the item and associative recognition thresholds (see Equation A10 and Equation A12 in Appendix A for more details). After making the judgment, the simulated subject updated the context with the test item or the second item in the test pair, following Equation 2 ($\beta = \beta_{cue}$). For items or pairs judged as old, the simulated subject updated the M^{FC} and M^{CF} matrices following Equation 1, to simulate output encoding. Additionally, context drifted between phases following Equation 2 ($\beta = \beta_{distract}$) and between study-test-test trials ($\beta = \beta_{post}^{recall}$). We repeated the study-test-test trials three times in one session, excluding the first trial from analysis, consistent with the real experiment. We conducted 300 simulation sessions using particle swarm optimization to identify parameter values that minimize the sum of squared deviations (Table 1).

Figure 12 illustrates simulation results alongside the experimental data. CMR-IA exhibited high correlations between repeated item recognition tests (the “Same Item” condition, $\mathcal{Q} = 0.78$) and repeated associative recognition tests (the “Intact Pair” condition, $\mathcal{Q} = 0.80$). The reinstatement of context during Test 1 ensures comparable recognition performance during Test 2. Furthermore, the mechanism of output encoding contributes to the slight increase in recognition performance when the same items or pairs are tested twice.

When testing different information, the value of $\mathcal{Q}$ for item recognition followed by associative recognition (the “Item/Pair” condition) is 0.58, while $\mathcal{Q}$ for associative recognition followed by item recognition (the “Pair/Item” condition) is 0.68. CMR-IA predicts a decrease in correlation compared to repeated tests of the same information due to the mechanistic differences between item and associative recognition (see a detailed analysis in Simulation 3). In addition, during item recognition, the process of output encoding minimally strengthens the association between the two items within a pair. Testing different items in a word pair reveals the weakest dependency relation (the “Diff

Item” condition, $Q = 0.27$). This occurs because the mnemonic information probed in Test 2 does not benefit from output encoding in Test 1.

In the “Repeated Lure” condition, a recognition lure from Test 1 reappears in Test 2. Despite neither item appearing in the study list, successive tests of these lures exhibit strong dependencies. CMR-IA captures this result with $Q = 0.58$, similar to that observed in the experiment. The model encodes lures incorrectly judged as old in Test 1, thereby increasing the probability of similar judgments in Test 2. Retrieval variability, controlled by the parameter σ , also plays a crucial role in preventing the correlation from becoming excessively high and ensuring alignment with the levels observed in the data.

6. General Discussion

We propose CMR-IA as a unified theory of memory for items and associations. Most prior developments of retrieved-context theory sought to explain recall dynamics in free recall. Recent papers have extended these models to account for data on item recognition (Healey & Kahana, 2016) and serial recall (Lohnas, 2025). The present paper fills a key gap by extending this class of models to data on cued recall and associative recognition and comparing memory for associations with memory for items, including novel comparisons of successive tests. Simultaneously accounting for data on the relations between item and associative memory has long constituted a key challenge for computational models of human memory (Murdock, 1992).

We adopted the Gestalt approach to associative learning by associating the two constituent items in a pair with the same context. This is realized by adding the two feature vectors together, updating the context with the combined feature vector, and storing the association in the associative matrices. Our approach ensures symmetry between the two items within a pair. An alternative approach would treat the pair as a two-item list, which updates the context sequentially for each item. Consequently, this alternative associates the two items with different contexts. Achieving associative symmetry under this approach requires more constraints on the parameter space, and it

remains an open question whether the model could maintain this symmetry while simultaneously simulating other memory phenomena. A hybrid approach is also possible, where one portion of the context remains stable across both items, serving as a shared context, while another portion varies with each item. Such a mechanism resembles existing frameworks that concatenate item features with associative features (Cox & Shiffrin, 2017; Cox, 2024), employ hierarchical representations (Osth & Hurlstone, 2023; Logan & Cox, 2023), or implement multi-scale contextual drift (Fox & Osth, 2023). Future research should investigate the necessity of this additional model complexity.

We designed a mechanism to exploit the stored associative information and demonstrate that it enables our model to perform associative recognition and cued recall. For associative recognition, one of the items in the test pair reinstates its associated context into the current context, and the model compares the updated context with the associated context of the other item. In cued recall, the cue item reinstates its associated context into the current context, and the model uses the updated context to retrieve the target. Unlike item recognition and free recall, where the model directly employs the current context, the additional step of reinstatement in both associative recognition and cued recall allows CMR-IA to effectively utilize stored associative information. Consequently, our model successfully simulates various benchmark phenomena in item and associative memory.

Item and Associative Recognition

Our first set of simulations aimed to evaluate whether CMR-IA provides a quantitative account of selected phenomena concerning item and associative recognition.

In the first simulation, we evaluated the model’s ability to account for the joint effects of recency and similarity on item recognition. Unable to locate an adequate archival dataset for secondary analyses, we conducted a novel experiment. Subjects performed a continuous recognition task, where we presented an item twice at varying lags and grouped same-category items into blocks during the presentation. As expected, our data revealed that subjects recognized recent items more accurately, particularly those with fewer similar

prior items. CMR-IA captures these effects through the interaction between contextual drift and pre-experimental associations between items (our model of similarity).

G. Schwartz et al. (2005) documented a contiguity effect in item recognition (see also Osth et al., 2018; Malmberg & Annis, 2012). They found that successively testing two old items on a recognition memory test increases the likelihood of high-confidence hits when items are studied in nearby list positions. Context reinstatement in CMR-IA produces this contiguity effect by retrieving a context similar to that associated with nearby studied items, thereby facilitating their recognition.

In contrast to the clear contiguity effect observed in item recognition, Osth and Fox (2019) found no evidence for such an effect in associative recognition. In their study, the false alarm rate for rearranged pairs was unaffected by the study lag between the items in the pair. Although CMR-IA theoretically predicts a decrease in false alarms for rearranged pairs as the lag increases, this effect is contingent on the model’s parameters. With the best-fitting parameters, the influence of lag is minimal and washed out by other factors.

Hockley (1992) demonstrated that items and associations exhibit different forgetting rates. Specifically, item recognition performance declines more rapidly than associative recognition performance as the study-test lag increases. CMR-IA accounts for this phenomenon by positing distinct mechanisms for item and associative recognition. The model stipulates that associative recognition includes an additional reinstatement step, which utilizes stored associative information and reduces dependence on the current temporal context, thereby being less influenced by the study-test lag.

Word frequency effects are well-documented effects in item recognition: rare words are more easily recognized as targets and more easily rejected as lures than common words (Schulman, 1967; Lohnas & Kahana, 2013). When simulating the word frequency effect in CMR-IA, we find that an attention mechanism and a threshold adjustment mechanism are both necessary. The attention mechanism enables CMR-IA to encode rare words more effectively during the study phase, thereby increasing the hit rate. On the other hand, the

model sets a higher recognition threshold for rare words during the test phase, which leads to a lower false alarm rate.

Cued Recall

We next evaluated CMR-IA’s ability to account for a set of key results concerning paired-associate (cued) recall. We began by examining whether our model could reproduce the classic serial position effect in cued recall. Specifically, empirical studies showed that the last couple of studied word pairs enjoy substantially higher recall accuracy than other word pairs, especially when tested immediately after study (Murdock, 1963). Consistent with its account in free recall, CMR-IA generates this recency effect through the evolving temporal context. The contexts associated with late-list items have greater similarity to the context at test, which boosts their recall probability, especially when the test is administered immediately after study.

We next considered the phenomenon of associative symmetry (Kahana, 2002). Whereas subjects in free recall exhibit a marked forward asymmetry in their recall transitions, this asymmetry disappears when testing paired associates. After studying a list of $A_i - B_i$ pairs, subjects can recall A_i given B_i just as easily as they can recall B_i given A_i . CMR-IA produces associative symmetry as a direct consequence of its symmetrical associative learning mechanism. Because A_i and B_i are encoded within the same context, providing either item as a cue retrieves this shared context. The retrieved context then serves as an equally effective cue for the other pair member.

Recall errors can provide valuable information about the nature of memory encoding and retrieval. In the case of cued recall, subjects will occasionally commit intrusions, which at times come from associations learned on prior lists (prior-list intrusions, or PLIs) or from other pairs within the current list (intra-list intrusions, or ILIs). Empirical data show that PLIs exhibit a marked recency effect, with intrusions more likely to come from recently studied lists. ILIs exhibit a contiguity effect, tending to originate from pairs studied in close temporal proximity to the target pair (Davis et al., 2008). CMR-IA

explains both phenomena through the contextual drift mechanism. The increased context similarity among recent lists accounts for the recency effect in PLIs, while the increased context similarity among nearby pairs accounts for the contiguity effect in ILIs.

Similarity can significantly influence both correct recall and errors in cued recall tasks. Specifically, the probability of correctly recalling a name is inversely related to the number of similar faces (i.e., neighbors) present in the study set. Furthermore, when ILIs occur, they are more likely to be names associated with faces that are similar to the cue face (Pantelis et al., 2008). CMR-IA naturally accounts for both of these empirical regularities. Within the model, similarity is instantiated as pre-experimental associations between items. Consequently, presenting a face cue strengthens the contextual representation of not just that face, but also its neighbors, proportional to their similarity. This process introduces competition at retrieval, which explains both the reduced accuracy for cues with many neighbors and the high probability that intrusions will come from similar faces.

Successive Tests of Item and Associative Information

To further explore the relation between item and associative information, we report two new experiments with accompanying CMR-IA fits. The first experiment, which uses successive tests of recognition and cued recall, reveals a higher correlation between associative recognition and cued recall than between item recognition and cued recall. The second experiment, utilizing successive recognition tests, demonstrates that greater informational overlap between tasks produces a higher correlation in performance.

Quantitative fits show that CMR-IA successfully accounts for the results of both experiments. The model explains the dissociation between item and associative information through its distinct retrieval mechanisms. Within the model, item information corresponds to the similarity between the current context and the context retrieved by the item. In contrast, associative information depends less on the current context and more on the strength of the learned association, which is represented by the similarity between the

two contexts retrieved by each member of the pair. Finally, by incorporating output encoding and retrieval variability, the model also captures more subtle dynamics of successive tests, such as the moderate contingency for repeated lures.

Relation to Other Models

Murdock's (1982) Theory of Distributed Associative Memory (TODAM) provided one of the first attempts at a theory of item and associative memory. As described in the Introduction, TODAM represented each studied item as an N-dimensional vector and assumed that learning involved the accumulation of these memories by vector addition. TODAM assumed that learning a novel pair of items involved the discrete (vector) convolution of the constituent items, and that summing the convolution along with the items enabled the storage of both items and associations in a common memory vector. TODAM simulated item and associative recognition as a similarity operation: If the dot-product between the probe item or association and the memory vector exceeded a threshold, the model would endorse the probe item as old; otherwise, the model would consider the probe item as new. Cued recall involved computing the inverse convolution of the probe with the memory vector to retrieve an approximate representation of the target. A hypothesized "deblurring" stage would convert the retrieved representation into a target memory, again using a similarity operation to determine the best match. In this manner, TODAM could simulate item recognition, associative recognition, and cued recall in a common framework. Concurrent with Murdock's TODAM, Metcalfe (Metcalfe-Eich, 1982) proposed a similar approach but used auto-convolution rather than direct summation as the basis for storing item information.

Subsequent years saw a gold rush of theory development. Gillund and Shiffrin (1984) extended their Search of Associative Memory model of free recall to account for data on item and associative information. Humphreys et al. (1989) developed a tensor model that allowed for three-way associations between context, items, and associations. Although list context did not directly play a role in Murdock and Metcalfe's original

implementations, most subsequent applications included a representation of context and in several cases, an autocorrelated representation of context (Murdock, 1997; Mensink & Raaijmakers, 1988). Following Murdock, most models assumed distinct storage of item and associative information (Cox, 2024; Osth & Dennis, 2015) and distinct retrieval operations for recognition and recall (Shiffrin & Steyvers, 1997; Osth & Dennis, 2015).

CMR-IA departs from most other theories by assuming a common learning rule for items and associations. Whether learning two neighboring items in a free recall task or learning a paired associate in a cued recall task, the model associates each item with a representation of context. However, in learning a pair of items, the model associates the same context with both items. In recognition and cued recall, the vector representing the cue item retrieves the history of previously associated context vectors. This retrieval operation, which sits at the heart of the model, drives both recognition and recall decisions. Because CMR-IA does not directly associate the constituent items of a paired associate, the model cannot use a direct similarity operation for associative recognition. Rather, the first item processed by the subject retrieves its associated context, which in turn updates the current context before comparing the retrieved context of the second item and the current context. Thus, the similarity between the retrieved context and the current context drives recognition decisions.

CMR-IA offers a unified account of item and associative information grounded in conjunctive encoding of items and context-sensitive similarity. Other recent models also unify item and associative recognition (Cox & Shiffrin, 2017; Cox, 2024; Osth & Dennis, 2015, 2024). However, CMR-IA goes further in that it encompasses both recall and recognition. It can therefore not only account for memory dynamics in both paradigms, but it can also explain key aspects of the relation between them, as measured by successive testing data. CMR-IA accomplishes this by combining the Gestalt assumption of associative learning with CMR’s recursive contextual updating and retrieval rules, and by expanding those rules to include learning during retrieval.

Limitations and Future Directions

To the extent that item and associative memory tasks share fundamental properties with tasks already modeled using CMR2 and its variants, our model’s success should not surprise anyone. Many fundamental principles of memory, including recency, similarity, contiguity, associative interference, and context-dependency, appear conserved across a wide range of tasks (Kahana, Diamond, & Aka, 2024). The data that will offer the strongest model tests and thus highlight potential limitations will come from dissociations among these tasks. For example, Simulation 3 showed that whereas CMR-IA naturally produced the differential forgetting rates of item and associative information, the model struggled to simultaneously produce the appropriate levels of performance in both tasks. Other dissociations between item and associative tasks come from studies of response times (Nobel & Shiffrin, 2001), and dynamic retrieval has played a key role in recent models of item and associative memory, enabling these models to account for both accuracy and response time distributions (Diller, Nobel, & Shiffrin, 2001; Cox, 2024). Our model, however, lacks a mechanism to account for response time in recognition memory tasks and thus cannot account for these data. Future work can change the decision mechanism to a racing diffusion model (Tillman, Van Zandt, & Logan, 2020) to incorporate a mechanism for response time.

The predicted differences between item and associative memory stem from the central assumption in our model: namely, the associative process that binds two items being encoded as a pair. We adopted the strong position that paired items associate with a single shared context vector. This allowed us to account for symmetric associations within pairs (Kahana, 2000) even though the associative process that links neighboring items in a free or serial recall experiment is inherently forward biased (Howard & Kahana, 2002). Our model did not provide a natural means of explaining the observed asymmetry in associative recognition.

When simulating the word frequency effect in item recognition (Simulation 4), we

introduced the attention mechanism and the threshold adjustment mechanism. We demonstrated that both are necessary for CMR-IA to produce the mirror effect. We implement the elevated attention for rare words by changing a single coefficient. However, this way is highly abstract, and a better approach would be to develop a detailed theory of attention and its interaction with memory. Moreover, we only incorporate this attention mechanism during the study of items. The analogous mechanism in associative learning is yet to be explored, and the word frequency effects in other tasks are yet to be examined. The multi-task dataset from Cox et al. (2020) can be a good testbed for future development.

More broadly, we acknowledge that in learning consistent pairings of items, subjects would likely create new conjunctive features. Compound words, such as “ice cream”, “pancake”, and “laptop”, naturally illustrate the creation of such conjunctive features and, ultimately, new units in memory. CMR-IA does not explain how new feature vectors emerge through the history of shared experience. We see this as a fundamental challenge for future work. Our work adhered to the common practice of fitting group average data rather than data from individual subjects. Thus, we only use one set of parameters to simulate each effect, mimicking the situation of an average subject performing many trials. This approach could not provide insights into individual differences in item and associative memory, including the correlations across subjects reported in our novel experimental data. A better approach would be to fit different sets of parameters for different subjects, thus accounting for the variability in responses. For example, Healey and Kahana (2016) fit individual young and old subjects in free recall tasks and showed that age-related memory change attributes to attention sustainability, context retrieval ability, intrusion screening ability, and retrieval noise. Cohen and Kahana (2022) and Lohnas (2025) also demonstrate the utility of individual subject fits in evaluating the predictions of memory theories.

Conclusions

The present work shows how a context-maintenance and retrieval model (CMR-IA) can explain a wide range of facts about item and associative memory. Our model captures associative effects by assuming that each item in a pair binds with an identical representation of context and context drifts between pairs and lists according to the standard CMR context evolution equation (Lohnas et al., 2015; Cohen & Kahana, 2022). The shared context among the items thus serves as the glue binding them together, symmetrically. As in free recall, context dynamics generate recency and contiguity at multiple time scales. Learning during recall combines with the core model mechanisms to explain the correlations between successive tests. Pre-experimental associations allow the model to account for similarity and interference effects. These findings demonstrate that retrieved-context theory readily extends from self-cued to experimenter-cued memory tasks.

References

- Anderson, J. R., & Bower, G. H. (1972). Recognition and retrieval processes in free recall. *Psychological Review*, 79(2), 97-123. doi: 10.1037/h0033773
- Asch, S. E., & Ebenholtz, S. M. (1962). The principle of associative symmetry. *Proceedings of the American Philosophical Society*, 106, 135-163.
- Bahrick, H. P. (1970). Two-phase model for prompted recall. *Psychological Review*, 77(3), 215-222. doi: 10.1037/h0029099
- Bates, D., Kliegl, R., Vasishth, S., & Baayen, H. (2018). Parsimonious mixed models. *arXiv*(arXiv:1506.04967). doi: 10.48550/arXiv.1506.04967
- Benjamin, A. S., Diaz, M., & Wee, S. (2009). Signal detection with criterion noise: applications to recognition memory. *Psychological review*, 116(1), 84. doi: 10.1037/a0014351
- Bower, G. H. (1972). Stimulus-sampling theory of encoding variability. In A. W. Melton & E. Martin (Eds.), *Coding processes in human memory* (pp. 85-121). New York: John Wiley and Sons.
- Caplan, J. B. (2005). Associative isolation: Unifying associative and list memory. *Journal of Mathematical Psychology*, 49, 383-402.
- Carr, H. A. (1931). The laws of association. *Psychological Review*, 38(3), 212-228. doi: 10.1037/h0075109
- Clark, S. E. (1992). Word frequency effects in associative and item recognition. *Memory & Cognition*, 20(3), 231-243.
- Cohen, R. T., & Kahana, M. J. (2022). A memory based theory of emotional disorders. *Psychological Review*, 129(4), 742-776. doi: 10.1037/rev0000334
- Cox, G. E. (2024). Dynamic retrieval of events and associations from memory: An integrated account of item and associative recognition. *Psychological Review*.
- Cox, G. E., & Shiffrin, R. M. (2017). A dynamic approach to recognition memory. *Psychological review*, 124(6), 795. doi: 10.1037/rev0000076

- Cox, G. E., & Shiffrin, R. M. (2024). Computational models of event memory. In M. J. Kahana & A. D. Wagner (Eds.), *Oxford handbook of human memory* (Vol. 1, pp. 64–93). Oxford University Press.
- Davis, O. C., Geller, A. S., Rizzuto, D. S., & Kahana, M. J. (2008). Temporal associative processes revealed by intrusions in paired-associate recall. *Psychonomic Bulletin & Review*, 15(1), 64–69.
- Diller, D. E., Nobel, P. A., & Shiffrin, R. M. (2001). An ARC–REM model for accuracy and response time in recognition and recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(2), 414–435. Retrieved from <http://dx.doi.org/10.1037/0278-7393.27.2.414> doi: 10.1037/0278-7393.27.2.414
- Ebbinghaus, H. (1885/1913). *Memory: A contribution to experimental psychology*. New York: Teachers College, Columbia University.
- Estes, W. K. (1955). Statistical theory of spontaneous recovery and regression. *Psychological Review*, 62, 145–154. doi: 10.1037/h0048509
- Flexser, A. J., & Tulving, E. (1978). Retrieval independence in recognition and recall. *Psychological Review*, 85, 153–171.
- Fox, J., & Osth, A. F. (2023). Modeling the continuous recognition paradigm to determine how retrieval can impact subsequent retrievals. *Cognitive Psychology*, 147, 1–35. doi: 10.1016/j.cogpsych.2023.101605
- Friendly, M., Franklin, P. E., Hoffman, D., & Rubin, D. C. (1982). The Toronto Word Pool: Norms for imagery, concreteness, orthographic variables, and grammatical usage for 1,080 words. *Behavior Research Methods and Instrumentation*, 14, 375–399. doi: 10.3758/BF03203275
- Gillund, G., & Shiffrin, R. M. (1984). A retrieval model for both recognition and recall. *Psychological Review*, 91(1), 1–67. doi: 10.1037/0033-295X.91.1.1
- Glanzer, M., Hilford, A., & Kim, K. (2004). Six regularities of source recognition. *Journal*

- of Experimental Psychology: Learning, Memory and Cognition*, 30(6), 1176–1195.
- Glenberg, A. M., & Bradley, M. M. (1979). Mental contiguity. *Journal of Experimental Psychology: Human Learning and Memory*, 5(2), 88-97.
- Glenberg, A. M., & Swanson, N. G. (1986). A temporal distinctiveness theory of recency and modality effects. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 12(1), 3-15. doi: 10.1037/0278-7393.12.1.3
- Gregg, V. (1976). Word frequency, recognition and recall. In J. Brown (Ed.), *Recall and recognition*. Oxford, England: John Wiley and Sons.
- Han, S., & Dobbins, I. G. (2008). Examining recognition criterion rigidity during testing using a biased-feedback technique: Evidence for adaptive criterion learning. *Memory & Cognition*, 36(4), 703-715.
- Healey, M. K., & Kahana, M. J. (2016). A four-component model of age-related memory change. *Psychological Review*, 123(1), 23–69. doi: 10.1037/rev0000015
- Herbart, J. F. (1834). *Lehrbuch zur Psychologie*. In K. Kehrbach (Ed.), *Sämtliche Werke* (2nd ed., Vol. 4, p. 376-377). Königsberg: August Wilhelm Unzer.
- Hirshman, E. (1995). Decision processes in recognition memory: Criterion shifts and the list-strength paradigm. *Journal of Experimental Psychology: Learning, Memory, & Cognition*, 21, 302-313.
- Hockley, W. E. (1982). Retrieval processes in continuous recognition. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 8, 497-512.
- Hockley, W. E. (1992). Item versus associative information: Further comparisons of forgetting rates. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18, 1321-1330.
- Hockley, W. E., & Murdock, B. B. (1987). A decision model for accuracy and response latency in recognition memory. *Psychological Review*, 94, 341-358.
- Holdstock, J. S., Mayes, A. R., Roberts, N., Cezayirli, E., Isaac, C. L., O'Reilly, R. C., & Norman, K. A. (2002). Under what conditions is recognition spared relative to recall

- after selective hippocampal damage in humans. *Hippocampus*, 12(3), 341–351. doi: 10.1002/hipo.10011
- Hollingsworth, H. L. (1928). *Psychology: Its facts and principles*. D. Appleton and Company.
- Hong, M. K., Gunn, J. B., Fazio, L. K., & Polyn, S. M. (2024). The modulation and elimination of temporal organization in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 50(7), 1035–1068. doi: 10.1037/xlm0001311
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79, 2554–2558.
- Howard, M. W., & Kahana, M. J. (1999). Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(4), 923–941. doi: 10.1037/0278-7393.25.4.923
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299. doi: 10.1006/jmps.2001.1388
- Humphreys, M. S. (1978). Item and relational information: A case for context independent retrieval. *Journal of Verbal Learning and Verbal Behavior*, 17, 175–187.
- Humphreys, M. S., Pike, R., Bain, J. D., & Tehan, G. (1989). Global matching: A comparison of the SAM, Minerva II, Matrix, and TODAM models. *Journal of Mathematical Psychology*, 33, 36–67.
- Kahana, M. J. (1993). Interactions between item, associative, and serial order information. *Unpublished doctoral dissertation*.
- Kahana, M. J. (2000). Contingency analyses of memory. In E. Tulving & F. I. M. Craik (Eds.), *Oxford handbook of human memory* (pp. 323–384). England: Oxford Press.
- Kahana, M. J. (2002). Associative symmetry and memory theory. *Memory & Cognition*, 30, 823–840.

- Kahana, M. J. (2012). *Foundations of human memory*. New York, NY: Oxford University Press.
- Kahana, M. J. (2020). Computational models of memory search. *Annual Review of Psychology*, 71(1), 107–138. doi: 10.1146/annurev-psych-010418-103358
- Kahana, M. J., & Caplan, J. B. (2002). Associative asymmetry in probed recall of serial lists. *Memory & Cognition*, 30(6), 841–849.
- Kahana, M. J., Diamond, N. B., & Aka, A. (2024). Laws of human memory. In M. J. Kahana & A. D. Wagner (Eds.), *Oxford handbook of human memory* (Vol. 1, pp. 29–63). Oxford University Press.
- Kahana, M. J., Lohnas, L. J., Healey, M. K., Aka, A., Broitman, A. W., Crutchley, P., . . . Weidemann, C. T. (2024). The Penn Electrophysiology of Encoding and Retrieval Study. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 50(9), 1421–1443. doi: 10.1037/xlm0001319
- Kahana, M. J., Rizzuto, D. S., & Schneider, A. (2005). Theoretical correlations and measured correlations: Relating recognition and recall in four distributed memory models. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 31(5), 933–953.
- Kinsbourne, M., & George, J. (1974). The mechanism of the word-frequency effect on recognition memory. *Journal of Verbal Learning and Verbal Behavior*, 13, 63–69.
- Kohler, W. (1940). *Dynamics in psychology*. New York: Liveright.
- Kragel, J. E., Ezzyat, Y., Sperling, M. R., Gorniak, R., Worrell, G. A., Berry, B. M., . . . Kahana, M. J. (2017). Similar patterns of neural activity predict memory function during encoding and retrieval. *NeuroImage*, 155, 60–71. doi: 10.1016/j.neuroimage.2017.03.042
- Logan, G. D. (2021). Serial order in perception, memory, and action. *Psychological Review*, 128(1), 1–44.
- Logan, G. D., & Cox, G. E. (2023). Serial order depends on item-dependent and

- item-independent contexts. *Psychological Review*. doi: 10.1037/rev0000422
- Lohnas, L. J. (2024). Free recall and memory search. In M. J. Kahana & A. D. Wagner (Eds.), *Oxford handbook of human memory* (Vol. 1, pp. 831–870). Oxford University Press.
- Lohnas, L. J. (2025). A retrieved context model of serial recall and free recall. *Computational Brain & Behavior*, 8(1), 1–35. doi: 10.1007/s42113-024-00221-9
- Lohnas, L. J., & Kahana, M. J. (2013). Parametric effects of word frequency in memory for mixed frequency lists. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39(6), 1943–1946. doi: 10.1037/a0033669
- Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363. doi: 10.1037/a0039036
- MacLeod, C. M., & Kampe, K. (1996). Word frequency effects on recall, recognition, and word fragment completion tests. *Journal of Experimental Psychology*, 22(1), 132–142.
- Malmberg, K. J., & Annis, J. (2012). On the relationship between memory and perception: sequential dependencies in recognition memory testing. *Journal of Experimental Psychology: General*, 141(2), 233. doi: 10.1037/a0025277
- Malmberg, K. J., & Nelson, T. O. (2003). The word frequency effect for recognition memory and the elevated-attention hypothesis. *Memory & Cognition*, 31, 35–43.
- Manning, J. R. (2024). Context reinstatement. In M. J. Kahana & A. D. Wagner (Eds.), *Oxford handbook of human memory* (Vol. 1, pp. 1114–1139). Oxford University Press.
- McGeoch, J. A. (1932). Forgetting and the law of disuse. *Psychological Review*, 39, 352–70. doi: 10.1037/h0069819
- Mensink, G.-J. M., & Raaijmakers, J. G. W. (1988). A model for interference and forgetting. *Psychological Review*, 95, 434–55. doi: 10.1037/0033-295X.95.4.434
- Metcalfe, J. (1985). Levels of processing, encoding specificity, elaboration, and CHARM. *Psychological Review*, 92, 1–38.

- Metcalfe-Eich, J. (1982). A composite holographic associative recall model. *Psychological Review*, 89, 627-661.
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arxiv preprint arXiv:1301.3781*. doi: 10.48550/arXiv.1301.3781
- Monaco, J. D., Abbott, L. F., & Kahana, M. J. (2007). Lexico-semantic structure and the word-frequency effect in recognition memory. *Learning and Memory*, 14(2), 204-13.
- Murdock, B. B. (1956). Backward learning in paired associates. *Journal of Experimental Psychology*, 51, 213-215.
- Murdock, B. B. (1963). Short-term retention of single paired associates. *Journal of Experimental Psychology*, 65, 433-443.
- Murdock, B. B. (1965). Associative symmetry and dichotic presentation. *Journal of Verbal Learning and Verbal Behavior*, 4, 222-226.
- Murdock, B. B. (1974). *Human memory: Theory and data*. Potomac, MD: Lawrence Erlbaum and Associates.
- Murdock, B. B. (1982). A theory for the storage and retrieval of item and associative information. *Psychological Review*, 89, 609-626.
- Murdock, B. B. (1992). Item and associative information in a distributed memory model. *Journal of Mathematical Psychology*, 36, 68-99.
- Murdock, B. B. (1997). Context and mediators in a theory of distributed associative memory (TODAM2). *Psychological Review*, 104, 839-862. doi: 10.1037/0033-295X.104.4.839
- Murdock, B. B. (1998). The mirror effect and attention-likelihood theory: A reflective analysis. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 24(2), 524-534.
- Murdock, B. B., & Kahana, M. J. (1993a). Analysis of the list strength effect. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 19(3), 689-697.

- Murdock, B. B., & Kahana, M. J. (1993b). List-strength and list-length effects: Reply to Shiffrin, Ratcliff, Murnane, and Nobel. *Journal of Experimental Psychology: Learning, Memory and Cognition*, *19*(6), 1450-1453.
- Murdock, B. B., & Lamon, M. (1988). The replacement effect: Repeating some items while replacing others. *Memory & Cognition*, *16*, 91-101.
- Nobel, P. A., & Shiffrin, R. M. (2001). Retrieval processes in recognition and cued recall. *Journal of Experimental Psychology: Learning Memory, and Cognition*, *27*, 384-413.
- Osgood, C. E. (1949). The similarity paradox in human learning. *Psychological Review*, *56*, 132-143.
- Osth, A. F., & Dennis, S. (2015). Sources of interference in item and associative recognition memory. *Psychological Review*, *122*(2), 260—311. doi: 10.1037/a0038692
- Osth, A. F., & Dennis, S. (2024). Global matching models of recognition memory. In M. J. Kahana & A. D. Wagner (Eds.), *Oxford handbook of human memory* (Vol. 1, pp. 895–922). Oxford University Press.
- Osth, A. F., & Fox, J. (2019). Are associations formed across pairs? a test of learning by temporal contiguity in associative recognition. *Psychonomic Bulletin & Review*, *26*(5), 1650–1656. doi: 10.3758/s13423-019-01616-7
- Osth, A. F., & Hurlstone, M. J. (2023). Do item-dependent context representations underlie serial order in cognition? commentary on logan (2021). *Psychological Review*, *130*(2), 513–545. doi: 10.1037/rev0000352
- Osth, A. F., Jansson, A., Dennis, S., & Heathcote, A. (2018). Modeling the dynamics of recognition memory testing with an integrated model of retrieval and decision making. *Cognitive psychology*, *104*, 106–142. doi: 10.1016/j.cogpsych.2018.04.002
- Pantelis, P. C., van Vugt, M. K., Sekuler, R., Wilson, H. R., & Kahana, M. J. (2008). Why are some people's names easier to learn than others? The effects of similarity on memory for face-name associations. *Memory & Cognition*, *36*(6), 1182–1195.
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009). A context maintenance and

- retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156. doi: 10.1037/a0014420
- Rabinowitz, J. C., Mandler, G., & Barsalou, L. W. (1977). Recognition failure: Another case of retrieval failure. *Journal of Verbal Learning and Verbal Behavior*, 16, 639–663.
- Rao, K. V., & Proctor, R. W. (1984). Study-phase processing and the word frequency effect in recognition memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 10, 386–394.
- Rhodes, M. G., & Jacoby, L. L. (2007). On the dynamic nature of response criterion in recognition memory: Effects of base rate, awareness, and feedback. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33, 305–320.
- Robinson, E. S. (1932). *Association theory to-day; an essay in systematic psychology*. New York: The Century Co.
- Schulman, A. I. (1967). Word length and rarity in recognition memory. *Psychonomic Science*, 9(4), 211–212. doi: 10.3758/BF03330834
- Schwartz, G., Howard, M. W., Jing, B., & Kahana, M. J. (2005). Shadows of the past: Temporal retrieval effects in recognition memory. *Psychological Science*, 16(11), 898–904. doi: 10.1111/j.1467-9280.2005.01634.x
- Schwartz, R. M., & Humphreys, M. S. (1974). Recognition and recall as a function of instructional manipulations of organization. *Journal of Experimental Psychology*, 102(3), 517–519. doi: 10.1037/h0035901
- Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912. doi: 10.1037/a0013396
- Shepard, R. N., & Teghtsoonian, M. (1961). Retention of information under conditions approaching a steady state. *Journal of Experimental Psychology*, 62, 302–309.
- Shiffrin, R. M., & Steyvers, M. (1997). A model for recognition memory: REM—retrieving

- effectively from memory. *Psychonomic Bulletin & Review*, 4(2), 145–166. doi: 10.3758/BF03209391
- Siegel, L. L., & Kahana, M. J. (2014). A retrieved context account of spacing and repetition effects in free recall. *Journal of Experimental Psychology: Learning Memory and Cognition*, 40(3), 755–764. doi: 10.1037/a0035585
- Solway, A., Murdock, B. B., & Kahana, M. J. (2012). Positional and temporal clustering in serial order memory. *Memory & Cognition*, 40(2), 177–190. doi: 10.3758/s13421-011-0142-8
- Stanislaw, H., & Todorov, N. (1999). Calculation of signal detection theory measures. *Behavior research methods, instruments, & computers*, 31(1), 137–149. doi: 10.3758/bf03207704
- Tillman, G., Van Zandt, T., & Logan, G. D. (2020). Sequential sampling models without random between-trial variability: The racing diffusion model of speeded decision making. *Psychonomic Bulletin & Review*, 27(5), 911–936. doi: 10.3758/s13423-020-01719-6
- Tulving, E. (1968). Theoretical issues in free recall. In T. R. Dixon & D. L. Horton (Eds.), *Verbal behavior and general behavior theory* (p. 2-36). Englewood Cliffs, N. J.: Prentice-Hall.
- Tulving, E., & Arbuckle, T. Y. (1963). Sources of intratrial interference in immediate recall of paired associates. *Journal of Verbal Learning and Verbal Behavior*, 1, 321–334.
- Tulving, E., & Thompson, D. M. (1973). Encoding specificity and retrieval processes in episodic memory. *Psychological Review*, 80, 352–373.
- Underwood, B. J. (1945). The effect of successive interpolations and proactive inhibition. *Psychological Monographs*, 59, v + 33. doi: 10.1037/h0093547
- Underwood, B. J., Boruch, R. F., & Malmi, R. A. (1978). Composition of episodic memory. *Journal of Experimental Psychology: General*, 107, 393–419.
- Usher, M., & McClelland, J. L. (2001). The time course of perceptual choice: The leaky,

- competing accumulator model. *Psychological Review*, 108(3), 550–592. doi: 10.1037/0033-295x.108.3.550
- Wachter, J. A., & Kahana, M. J. (2024, 10). A retrieved-context theory of financial decisions. *The Quarterly Journal of Economics*, 139(2), 1095–1147. doi: 10.1093/qje/qjad050
- Wallace, W. (1978). Recognition failure of recallable words and recognizable words. *Journal of Experimental Psychology: Human Learning and Memory*, 4, 441–452.
- Weidemann, C. T., Kragel, J. E., Lega, B. C., Worrell, G. A., Sperling, M. R., Sharan, A. D., . . . Kahana, M. J. (2019). Neural activity reveals interactions between episodic and semantic memory systems during retrieval. *Journal of Experimental Psychology: General*, 148(1), 1–12. doi: 10.1037/xge0000480
- Wickelgren, W. A., & Norman, D. A. (1966). Strength models and serial position in short-term recognition memory. *Journal of Mathematical Psychobiology*, 3, 316–347.
- Yang, J., Zhao, P., Zhu, Z., Mecklinger, A., Fang, Z., & Li, H. (2013). Memory asymmetry of forward and backward associations in recognition tasks. [doi:10.1037/a0028875]. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39(1), 253–269. doi: 10.1037/a0028875

Appendix A

Formal Description of CMR-IA

Structure and Initialization

Inheriting from CMR2 (Lohnas et al., 2015; Healey & Kahana, 2016), CMR-IA has two representational layers, each defined as a vector: the feature layer F and the context layer C . The i^{th} item or pair presented to the model activates its associated features in F and is denoted by the vector $\mathbf{f}_i$. This in turn updates the state of context to the vector $\mathbf{c}_i$, as described below. Each of these vectors is initialized to 0.

The layers interact through two associative matrices: M^{FC} , which stores the strengths of the feature-to-context associations, and M^{CF} , which stores the strength of the context-to-feature associations. Each association matrix is a weighted sum of a pre-experimental component (M_{pre}^{FC} and M_{pre}^{CF}) that reflects longstanding semantic relationships and an experimental component (M_{exp}^{FC} and M_{exp}^{CF}) that reflects new learning occurring during the experiment. At the start of each experimental session, the experimental associations are initialized to 0. The semantic associations in M_{pre}^{FC} and M_{pre}^{CF} are defined using Word2Vec, which is a method of estimating the representations of words in vector space, or word “embeddings” (Mikolov, Chen, Corrado, & Dean, 2013). The cosine of the angle between two words’ vector representations serves as the measure of semantic similarity. Thus, each element (a, b) in M_{pre}^{FC} and M_{pre}^{CF} is determined by taking the cosine similarity value between words a and b (though here we define an item’s similarity with itself as 0).

Item Presentation

As in CMR2, we assume that each item has a localist representation for simplicity (Lohnas et al., 2015; Healey & Kahana, 2016). The first item presented in each simulated session is not a list item, but rather a distraction item used to initialize the state of the context vector to have nonzero elements. Each $\mathbf{f}_i$ retrieves the context states to which

those features have previously been associated as the input to C :

$$\mathbf{c}_i^{\text{IN}} = \frac{M^{FC} \mathbf{f}_i}{\|M^{FC} \mathbf{f}_i\|}. \quad (\text{A1})$$

This retrieved context, $\mathbf{c}_i^{\text{IN}}$, which is normalized to have a length of 1 so that it can be directly compared to other states of context, is incorporated into the context representation by adding it to the current context vector $\mathbf{c}_{i-1}$:

$$\mathbf{c}_i = \rho_i \mathbf{c}_{i-1} + \beta \mathbf{c}_i^{\text{IN}}. \quad (\text{A2})$$

Note that Equation A1 and Equation A2 are equivalent to Equation 2 in the main text. In this way, context is a recency-weighted sum of past context states. During item encoding, $\beta = \beta_{enc}$ determines how much new information ($\mathbf{c}_i^{\text{IN}}$) is added into context with each studied item. ρ_i weakens the previous state of context such that $\|\mathbf{c}_i\| = 1$:

$$\rho_i = \sqrt{1 + \beta^2 [(\mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}})^2 - 1]} - \beta (\mathbf{c}_{i-1} \cdot \mathbf{c}_i^{\text{IN}}). \quad (\text{A3})$$

Pair Presentation

In simulations of cued recall and associative recognition, pairs of items are presented. Suppose the two items in a pair are represented as $\mathbf{f}_{i_1}$ and $\mathbf{f}_{i_2}$, respectively. The model adds them together to form a unified vector:

$$\mathbf{f}_i = \mathbf{f}_{i_1} + \mathbf{f}_{i_2}. \quad (\text{A4})$$

The model then follows Equation A1 and Equation A2 to update the context.

Forming Associations Between Items and Context

As each new item or pair is presented, new experimental associations are formed between the feature representation and the current state of context. These associations are

formed according to a Hebbian outer-product learning rule:

$$\begin{aligned}\Delta M_{exp}^{FC} &= \mathbf{c}_i \mathbf{f}_i^\top \\ \Delta M_{exp}^{CF} &= \mathbf{f}_i \mathbf{c}_i^\top.\end{aligned}\tag{A5}$$

We argue that our definition of association, which we inherit from Polyn et al. (2009), is more plausible in associative learning, compared to CMR2 (Lohnas et al., 2015; Healey & Kahana, 2016), which updates the associative matrices using $\mathbf{c}_{i-1}$ rather than $\mathbf{c}_i$. In our definition, $\mathbf{c}_i$ incorporates information from both $\mathbf{f}_{i_1}$ and $\mathbf{f}_{i_2}$, facilitating the formation of associations between the two items. In contrast, $\mathbf{c}_{i-1}$ lacks any information about either item. Consequently, when later cuing with $\mathbf{f}_{i_1}$, the context input encompasses both $\mathbf{c}_{i-1}$ and $\mathbf{f}_{i_1}$. Here, only $\mathbf{c}_{i-1}$ aids the retrieval of $\mathbf{f}_{i_2}$, while $\mathbf{f}_{i_1}$ fails to contribute but instead misleads the model toward retrieving items in the next pair. This presents challenges in cued recall.

The newly formed experimental associations are incorporated with pre-experimental semantic associations. The balance between new and existing associations is controlled by parameters γ_{FC} and γ_{CF} :

$$\begin{aligned}M^{FC} &= (1 - \gamma_{FC})(I + s_{FC}M_{pre}^{FC}) + \gamma_{FC}\psi_i M_{exp}^{FC} \\ M^{CF} &= (1 - \gamma_{CF})(I + s_{CF}M_{pre}^{CF}) + \gamma_{CF}\phi_i M_{exp}^{CF},\end{aligned}\tag{A6}$$

where s_{FC} and s_{CF} are scaling parameters that control the influence of pre-experimental semantic associations. I is an identity matrix the same size as M_{pre}^{CF} and M_{pre}^{FC} . Effectively, this means that the on-diagonal terms are not multiplied by the s parameter. This allows the s parameter to scale semantic relations between pairs of different items while having no effect on auto-associations.

ϕ_i simulates increased attention to beginning-of-list items, producing a primacy

effect by scaling the magnitude of context-to-feature associations:

$$\phi_i = \phi_s e^{-\phi_d(i-1)} + 1, \quad (\text{A7})$$

where ϕ_s is a model parameter that scales the overall level of primacy, and the model parameter ϕ_d scales the degree to which primacy decays for each list item presented after the first item.

ψ_i simulates increased attention to rare words by scaling the magnitude of feature-to-context associations. Instead of adding a direct representation of word frequency into the model, we use a word's average pre-experimental semantic associations with other words ($\mathbf{s}_i^{\text{mean}}$) as the indicator of word frequency, since we have found a negative linear relationship between log word frequency and $\mathbf{s}_i^{\text{mean}}$ (see Simulation 4). We assume a linear relationship between the increased attention and $\mathbf{s}_i^{\text{mean}}$:

$$\psi_i = \psi_s \mathbf{s}_i^{\text{mean}} + \psi_c, \quad (\text{A8})$$

where ψ_s is a model parameter for slope and ψ_c is a model parameter for intercept. ψ_s is above 0 so that a rare word (with a high $\mathbf{s}_i^{\text{mean}}$) corresponds to a high ψ_i . Note that this mechanism is only necessary for simulating the word frequency effect in Simulation 4. For other simulations, we set $\psi_s = 0$ and $\psi_c = 1$ so that $\psi_i = 1$ for all items, thus being consistent with previous CMR models.

The Item Recognition Process

During item recognition, the test probe, represented as $\mathbf{f}_i$, is presented and it retrieves its associated context $\mathbf{c}_i^{\text{IN}}$ following Equation A1. The current state of context $\mathbf{c}_t$ is used to compare with $\mathbf{c}_i^{\text{IN}}$:

$$r_t^{\text{item}} = \mathbf{c}_i^{\text{IN}} \cdot \mathbf{c}_t. \quad (\text{A9})$$

For old items, the match between the retrieved context and the current context

depends on how much contextual drift has occurred between the original presentation and the recognition event. If the comparison returns r_t^{item} that is above a threshold $c_{thresh,t}^{item}$, the probe is judged as an old item, otherwise, it is rejected as a new item. Since the expected context similarity might vary during an experiment, to allow for dynamic adjustment of the recognition threshold, we propose that $c_{thresh,t}^{item}$ is derived from an exponentially weighted average of the previous ten items' context similarities $\bar{r}_t^{item}$ ¹⁶ and scaled by a factor a_i . Moreover, participants might make adjustments to the recognition threshold by setting a high threshold for saying “yes” to a rare word, as they may want to strike a balance between increasing their hit rate and lowering their false alarm rate. Thus, we assume a linear relationship between a_i and $\mathbf{s}_i^{\text{mean}}$:

$$\begin{aligned}\bar{r}_t^{item} &= \frac{1}{\sum_{k=t-10}^{t-1} e^{c_d(10-t+k)}} \sum_{k=t-10}^{t-1} e^{c_d(10-t+k)} r_k^{item} \\ a_i &= c_s \mathbf{s}_i^{\text{mean}} + c_{recog}^{item} \\ c_{thresh,t}^{item} &= a_i \bar{r}_t^{item} + \epsilon.\end{aligned}\tag{A10}$$

c_d is a model parameter for decay rate. c_s is a model parameter for slope and c_{recog}^{item} is a model parameter for intercept. c_s is positive so that a rare word (with a high $\mathbf{s}_i^{\text{mean}}$) corresponds to a high $c_{thresh,t}^{item}$. Such a relationship between the recognition threshold and the word frequency is only necessary for simulating the word frequency effect in Simulation 4. Modeling retrieval variability, ϵ is a random value drawn from a uniform distribution with an interval of $[-\sigma, \sigma]$, where σ is a model parameter. This mechanism of retrieval variability is only necessary for simulating Experiment 3. For other simulations, we set $c_s = 0$ and $\sigma = 0$ so that $c_{thresh,t}^{item} = c_{recog}^{item} \bar{r}_t^{item}$.

After the judgment, no matter what the judgment is, the context updates following

¹⁶ In implementation, the model maintains a queue of ten previous context similarities. To initialize the queue, we fill it with the average context similarity of five randomly selected studied items and five randomly selected non-studied items. The initialization takes place at the start of the test phase in a study-test paradigm, while in a continuous recognition paradigm, the initialization takes place at the first trial where it is possible for a studied item to reappear.

Equation A2, with a drift rate parameter for the cue, β_{cue} . If output encoding is on, the model also updates M^{FC} and M^{CF} for judged-as-old items following Equation A5.

The Associative Recognition Process

During associative recognition, a pair of test probes, represented as $\mathbf{f}_{i_1}$ and $\mathbf{f}_{i_2}$, is presented. The model first updates the current state of context with $\mathbf{f}_{i_1}$ following Equation A1 and Equation A2 ($\beta = \beta_{cue}$). Then, the updated context vector $\mathbf{c}'_t$ is used to compare with the context $\mathbf{c}_{i_2}^{\text{IN}}$ retrieved by $\mathbf{f}_{i_2}$ following Equation A1:

$$r_t^{\text{assoc}} = \mathbf{c}_{i_2}^{\text{IN}} \cdot \mathbf{c}'_t. \quad (\text{A11})$$

Because two items in a studied pair are associated with the same context that prevailed when they were originally presented, the retrieved contexts of these two items would be similar. Thus, the updated context (incorporating the retrieved context of the first item) should have a high similarity value with the retrieved context of the second item for an old (intact) pair, but a low similarity value for a new (rearranged) pair. If the comparison returns r_t^{assoc} that is above a threshold $c_{\text{thresh},t}^{\text{assoc}}$, then the test pair is judged as an old (intact) pair, otherwise, it is rejected as a new (rearranged) pair. Similar to item recognition, $c_{\text{thresh},t}^{\text{assoc}}$ is derived from an exponentially weighted average of the previous ten pairs' context similarities $\bar{r}_t^{\text{assoc}}$ ¹⁷ and scaled by the model parameter $c_{\text{recog}}^{\text{assoc}}$:

$$\begin{aligned} \bar{r}_t^{\text{assoc}} &= \frac{1}{\sum_{k=t-10}^{t-1} e^{c_d(10-t+k)}} \sum_{k=t-10}^{t-1} e^{c_d(10-t+k)} r_k^{\text{assoc}} \\ c_{\text{thresh},t}^{\text{assoc}} &= c_{\text{recog}}^{\text{assoc}} \bar{r}_t^{\text{assoc}} + \epsilon. \end{aligned} \quad (\text{A12})$$

c_d is a model parameter for decay rate. ϵ is a random value drawn from a uniform distribution with an interval of $[-\sigma, \sigma]$ representing retrieval variability. Again, this

¹⁷ At the start of the test, $\bar{r}_t^{\text{assoc}}$ is initialized by computing the average context similarity of five randomly selected studied pairs and five randomly selected new or rearranged pairs, depending on the experimental design. In a continuous recognition paradigm, this initialization takes place at the first trial where it is possible for studied pairs to reappear.

mechanism of retrieval variability is only necessary for simulating Experiment 3. For other simulations, we set $\sigma = 0$ so that $c_{thresh,t}^{assoc} = c_{recog}^{assoc} \bar{r}_t^{assoc}$.

After the judgment, no matter what the judgment is, $\mathbf{c}_{i_2}^{\mathbf{IN}}$ updates the state of context following Equation A2 with β_{cue} . If output encoding is on, the model also updates M^{FC} and M^{CF} for judged-as-old pairs following Equation A4 and Equation A5.

The Cued Recall Process

During cued recall, the cue item, represented as $\mathbf{f}_i$, is presented and it updates the current state of context following Equation A1 and Equation A2 ($\beta = \beta_{cue}$). Then, the updated contextual state $\mathbf{c}'_t$ is used for retrieval via the M^{CF} associations, similar to the free recall process in CMR2 (Lohnas et al., 2015; Healey & Kahana, 2016):

$$\mathbf{f}_t^{\mathbf{IN}} = M^{CF} \mathbf{c}'_t. \quad (\text{A13})$$

The resulting $\mathbf{f}_t^{\mathbf{IN}}$ gives the degree of support, or activation, for each item in the model's vocabulary. Items with low activation values are unlikely to be recalled by the retrieval process described below, and considering them as candidates for retrieval is extremely computationally expensive. Therefore, only a certain number of items¹⁸ with the highest activation values are assigned to a vector, $\mathbf{a}$, of retrieval candidates. The values in $\mathbf{a}$ are then used as the initial input for a set of competitive accumulators, one for each candidate, according to the leaky competitive accumulator model (Usher & McClelland, 2001):

$$\begin{aligned} \mathbf{x}_n &= (1 - \tau\kappa - \tau\lambda\mathbf{N})\mathbf{x}_{n-1} + \tau\mathbf{a} + \epsilon \\ \mathbf{x}_n &\rightarrow \max(\mathbf{x}_n, 0), \end{aligned} \quad (\text{A14})$$

¹⁸ Typically this number is four times the number of items in a study list. The two exceptions are that in Simulation 8 this number is 16 as only 16 items occur in one simulation session and that in Simulating Experiment 2 this number is 176, which is two times the number of items in the study list, as the study list is too long.

where $\mathbf{x}_n$ is a vector with one element for each retrieval candidate in $\mathbf{a}$. When the retrieval competition starts, all elements are set to zero (i.e., $\mathbf{x}_0 = 0$) and the activation for each item given in $\mathbf{a}$ is used as its starting line in the race to threshold. τ is a fixed time constant. κ is a parameter that determines the decay rate for item activations. λ is the lateral inhibition parameter, scaling the strength of an inhibitory matrix $\mathbf{N}$ that subtracts each item’s activation from all of the others except itself. ϵ is a random vector whose elements are drawn from a normal distribution with mean 0 and SD as a model parameter η . The second line of Equation A14 means that the accumulating elements cannot take on negative values. $\mathbf{x}_n$ continues to be updated until one of the activation values exceeds its threshold or until the recall period ends.

Similar to CMR2, CMR-IA dynamically sets the retrieval threshold of each item as the recall period progresses to allow items that were recalled earlier in the period to participate in, but not dominate, current retrieval competitions (Lohnas et al., 2015; Healey & Kahana, 2016). Specifically, at the beginning of the recall period, each item, i , has a threshold of $\Theta_i = 1$. If item i is retrieved, its threshold is incremented by a value ω and then gradually returns to 1 with subsequent recalls:

$$\Theta_i = 1 + \omega\alpha^j, \tag{A15}$$

where j is the number of subsequent retrievals, α is a parameter between 0 and 1; the larger the value of α , the more intervening retrievals are needed before an already-recalled item is likely to be retrieved again. Using this same mechanism, CMR-IA prevents the cue item from being retrieved by raising the threshold of the cue item following Equation A15 immediately after the presentation of the cue.

The first word to accumulate enough activation to cross its threshold wins the retrieval competition. The winner’s representation is reactivated on F , allowing the model to retrieve the contextual state associated with the item. Context is updated with the

same mechanism used during the study period, following Equation A2 with β_{rec} . If output encoding is on, the model also encodes the cue and the retrieved item as a pair by updating M^{FC} and M^{CF} following Equation A4 and Equation A5.

As in CMR2, before the item is actually outputted by the model, however, it undergoes a post-retrieval editing phase, consistent with the observation that people often report thinking of items that they do not overtly recall during free recall experiments. Editing is accomplished by comparing the context representation retrieved by the candidate item with the currently active context representation, similar to Equation A9. If the comparison returns a value that is beneath a threshold parameter, c_{thresh} , the item is filtered out.

Shift Between Different Phases

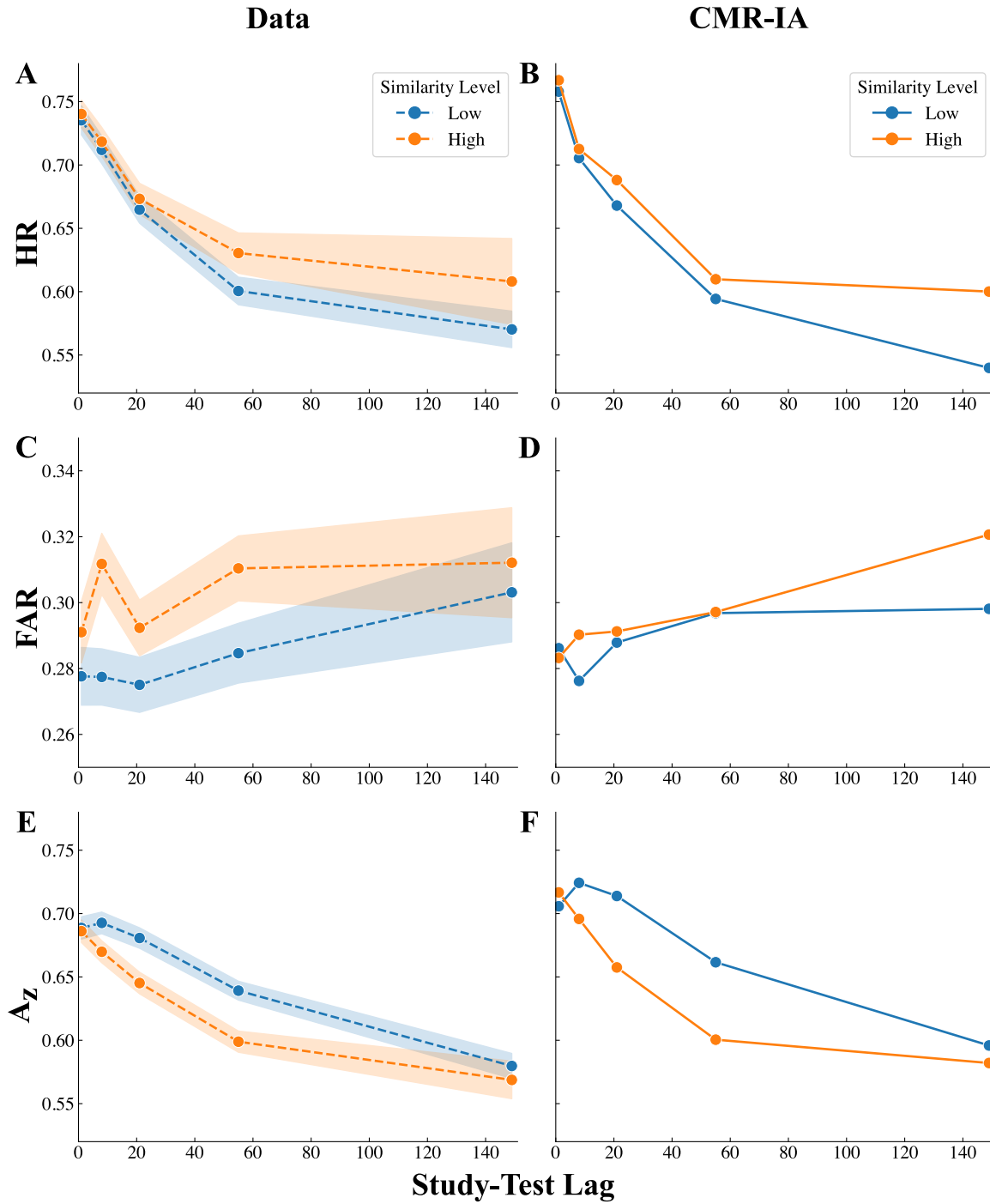
It is possible that the temporal context shifts when switching between different phases in an experiment. Some experiments even exaggerate this shift by adding a distraction task between studying and testing. CMR-IA simulates this shift in temporal context by activating a unique “disruption” item on the feature layer and allowing this item to update context using Equation A2, with a drift rate parameter $\beta_{distract}$ when shifting from a study phase to a test phase and another drift rate parameter β_{post}^{recall} when ending a test phase.

Category	Parameter	1	2a	2b	3	4	5	6a	6b	7	8	Exp 2	Exp 3
CMR Basic	β_{enc}	0.012	0.258	0.165	0.494	0.200	0.867	0.866	0.745	0.846	0.761	0.614	0.864
	β_{rec}						0.378	0.652	0.522	0.083	0.825	0.611	
	β_{cue}		0.288	0.504	0.378	0.920	0.650	0.426	0.958	0.693	0.998	0.987	0.209
	$\beta_{distract}$		0.407	0.452		0.370	0.749	0.067	0.131	0.220	0.850	0.409	0.365
	β_{recall}	0.146	0.030	0.595	0.682		0.819	0.173	0.414	0.224		1.000	0.952
Semantics	γ_{FC}	0.275	0.072	0.046	0.931	0.588	0.569	0.182	0.652	0.776	0.497	0.445	0.101
	γ_{CF}						0.561	0.206	0.534	0.488	0.735	0.701	
	s_{FC}	0.283	0.758	0.001	0.034	0.011	0.013	0.060	0.029	0.303	0.280	0.060	0.106
Primacy	s_{CF}						0.631	0.499	0.921	0.347		0.218	
	ϕ_s						0.198	5.981	5.094	1.496	7.423	2.424	
	ϕ_d						0.638	0.036	2.849	0.013	0.419	3.043	
Accumulator	κ						0.078	0.359	0.047	0.050	0.025	0.428	
	λ						0.171	0.071	0.135	0.147	0.184	0.023	
	η						0.025	0.039	0.026	0.022	0.050	0.147	
Threshold	ω						1.842	5.350	7.398	7.915	3.947	8.623	
	α						0.627	0.650	0.816	0.961	0.676	0.534	
	c_{thresh}	1.001			1.027	-0.395	0.251	0.151	0.118	0.364	0.180	0.181	
Attention	c_{item}^{recog}				0.986							1.165	1.090
	c_{assoc}^{recog}											1.025	1.108
	c_d	8.218	5.172	6.279	2.388	3.316						2.818	0.285
	c_s					11.590							
	σ												0.014
Attention	ψ_s					55.769							
	ψ_c					-4.156							

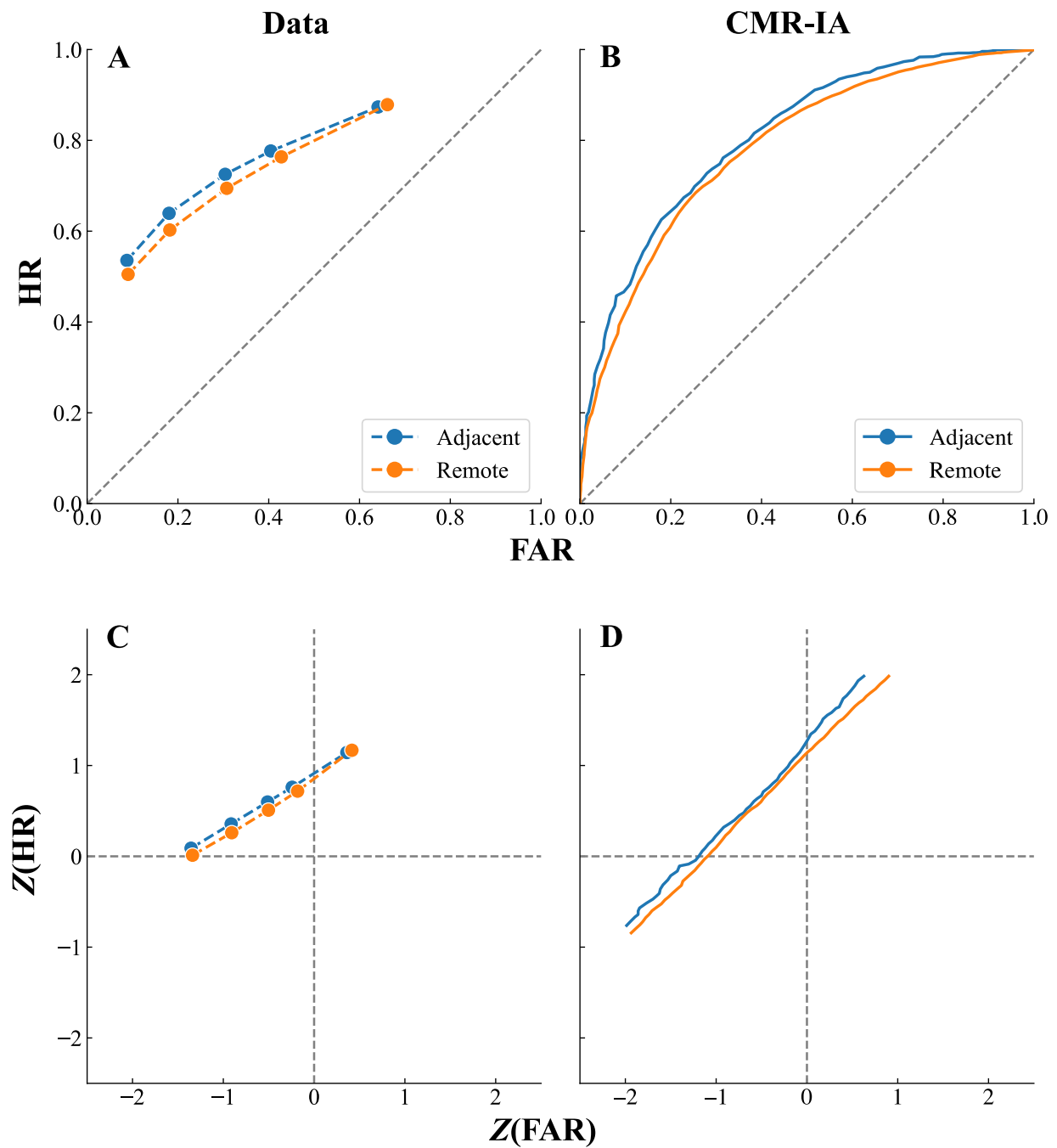
Table 1
Best-fit parameters of CMR-1A.

Congruent					Incongruent				
Data	Test 2	Test 1			Test 2	Test 1			
			+	-			+	-	
		+	0.319	0.012		+	0.293	0.122	
		-	0.006	0.663		-	0.049	0.537	
		Yule's $\mathcal{Q} = 0.94$				Yule's $\mathcal{Q} = 0.96$			
CMR-IA	Test 2	Test 1			Test 2	Test 1			
			+	-			+	-	
		+	0.300	0.033		+	0.304	0.122	
		-	0.025	0.642		-	0.020	0.554	
		Yule's $\mathcal{Q} = 0.98$				Yule's $\mathcal{Q} = 0.94$			

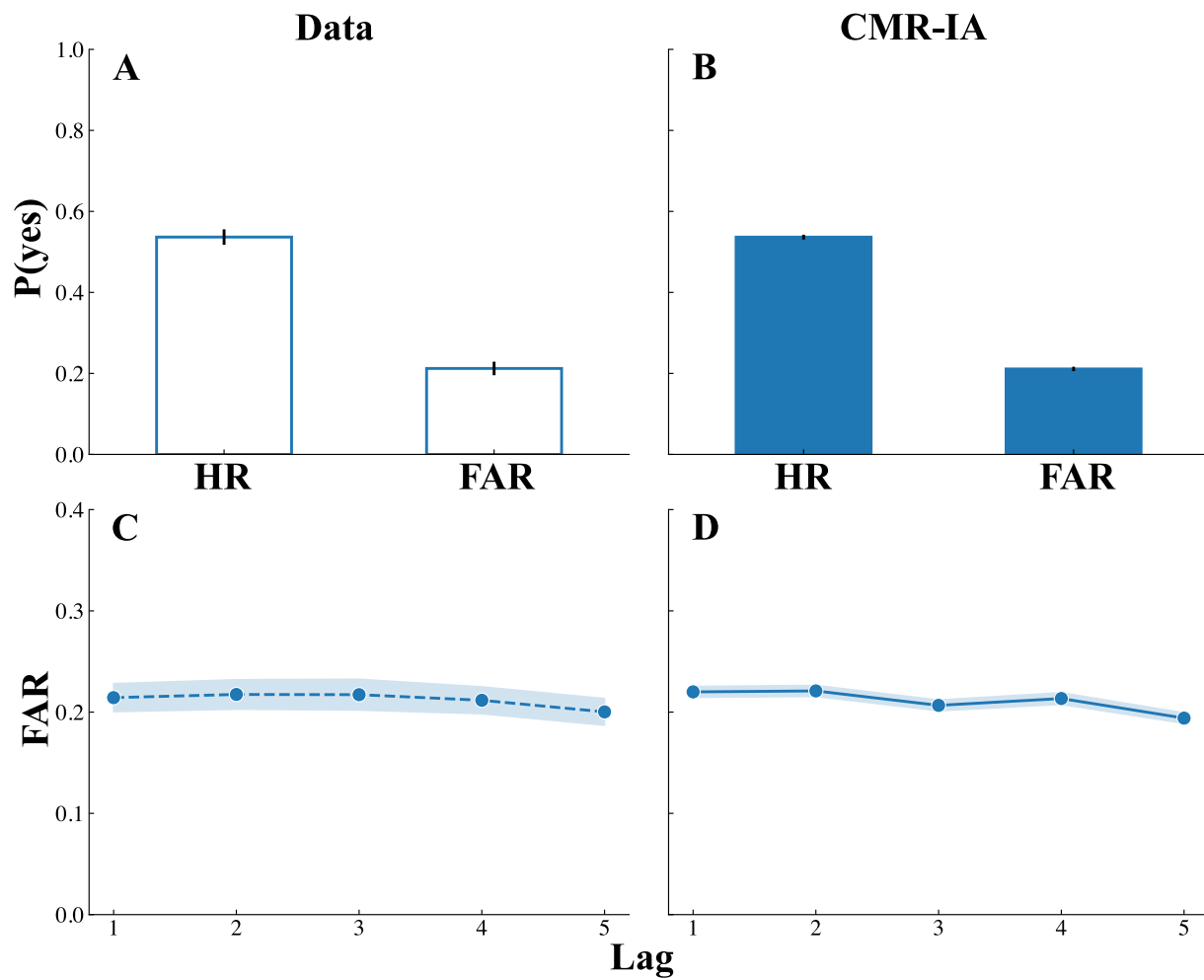
Table 2
Contingency tables for congruent and incongruent successive tests in Kahana (2002) and in CMR-IA.

**Figure 1**

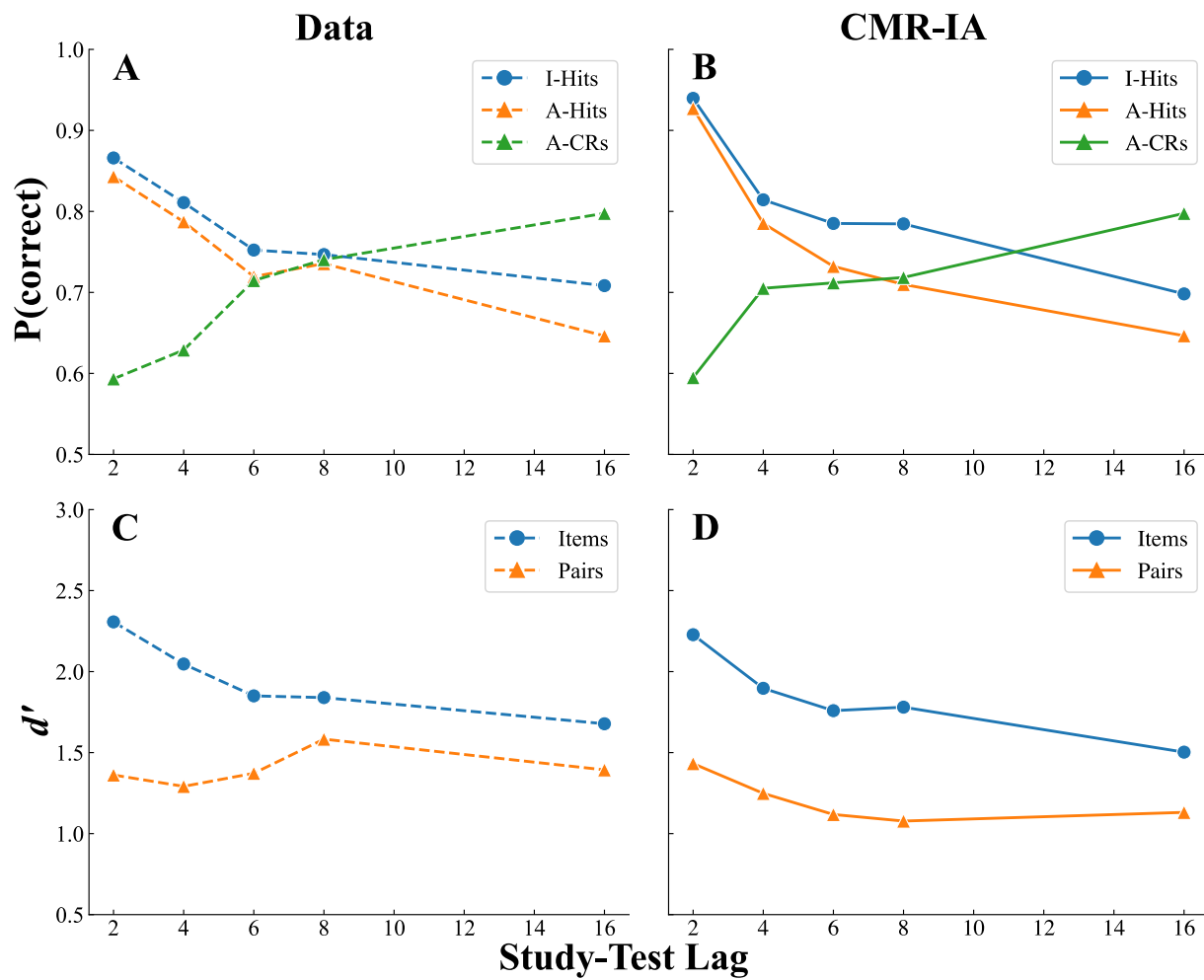
Joint effects of recency and similarity in continuous recognition. A high similarity level indicates the presence of at least two same-category items within the previous eight items, while a low similarity level indicates the presence of zero or one same-category item within the previous eight items. In Experiment 1, we observed the hit rate (HR), false alarm rate (FAR), and A_z as a function of both similarity and recency (panels A, C, E). CMR-IA predicts similar outcomes (panels B, D, F). Error bars indicate the standard error of the mean.

**Figure 2**

Successive-probe contiguity effects in item recognition. We compared test probes preceded by an old-item probe studied in an adjacent position (Adjacent) with those preceded by an old-item probe studied in a remote position (Remote). The receiver operating characteristic (ROC) curve and the z-ROC curve illustrate hit rates (HR) as a function of local false alarm rates (FAR). G. Schwartz et al. (2005) observed superior recognition performance for Adjacent probes (panels A, C). CMR-IA predicts comparable outcomes (panels B, D).

**Figure 3**

A null effect of temporal distance in associative recognition. Constructing rearranged pairs by combining items from pairs studied at varying list positions, Osth and Fox (2019) found no reliable evidence of contiguity across lags of one to five pairs. While matching the overall hit rate (HR) and false alarm rate (FAR) from Osth and Fox (panels A, B), CMR-IA yields a negligible relation between FARs and temporal distance (lag) at study (panels C, D). Error bars indicate the standard error of the mean.

**Figure 4**

Differential forgetting rate for item and associative recognition. Hockley (1992) reported hit rates (Hits) and correct rejection rates (CRs) for associative recognition (A) and item recognition (I) as a function of study-test lags (panel A). He demonstrated that item information declines more rapidly than associative information, as evidenced by d' values (panel C). CMR-IA predicts similar outcomes (panels B, D).

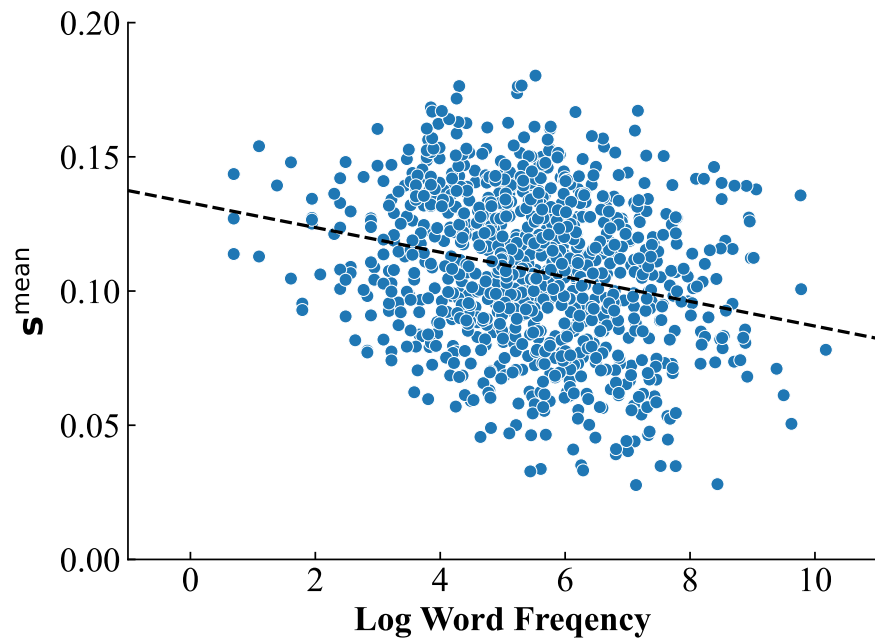
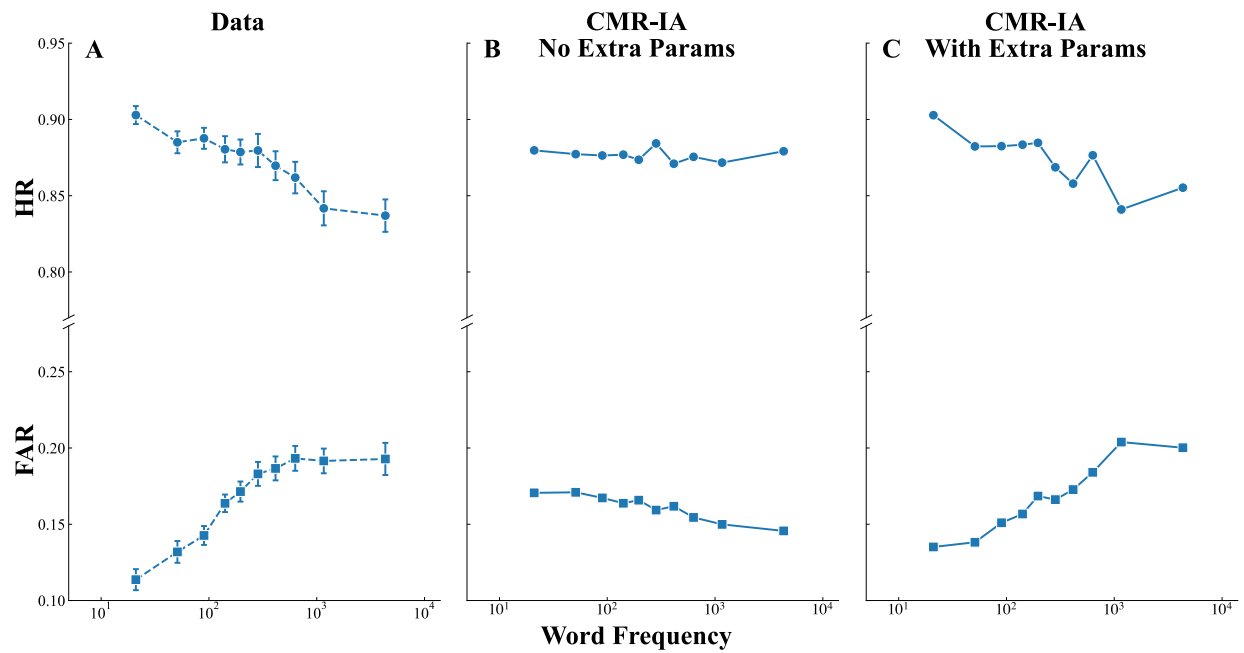


Figure 5

We observed a negative correlation between a word's average semantic associations with other words, s^{mean} , and its log word frequency among the 984 words used by Lohnas and Kahana (2013). Semantic associations were computed using Word2Vec.

**Figure 6**

Word frequency effects. As shown by Lohnas and Kahana (2013), the hit rate (HR) decreases while the false alarm rate (FAR) increases with rising word frequency (panel A). Initially, CMR-IA fails to exhibit the word frequency effect (panel B). After incorporating the attention and threshold adjustment mechanisms, CMR-IA accurately predicts the word frequency effect (panel C).

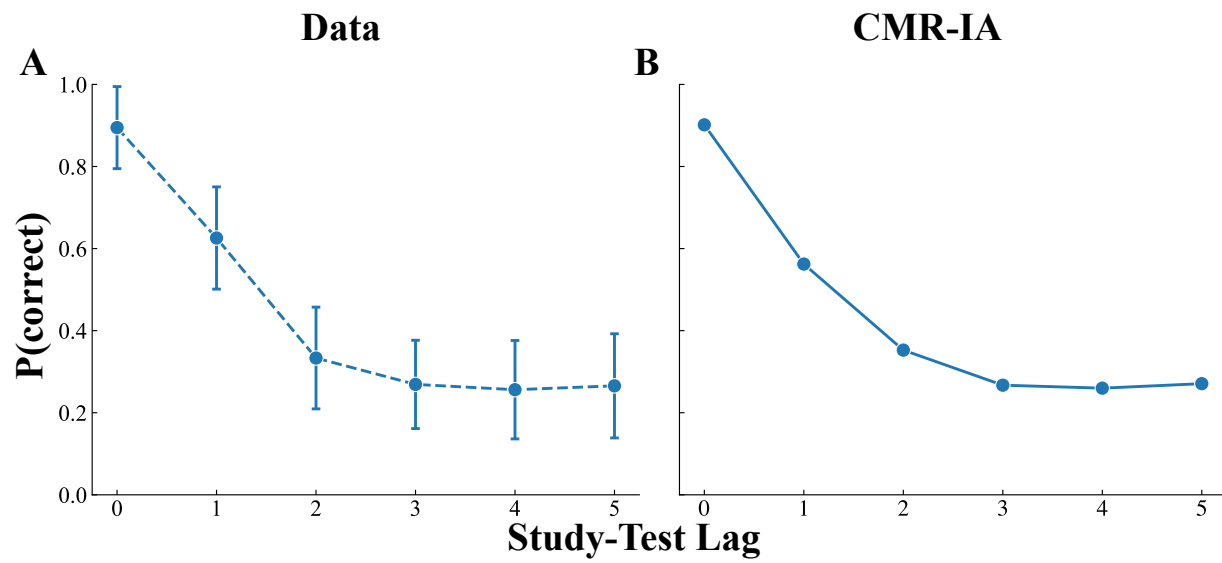
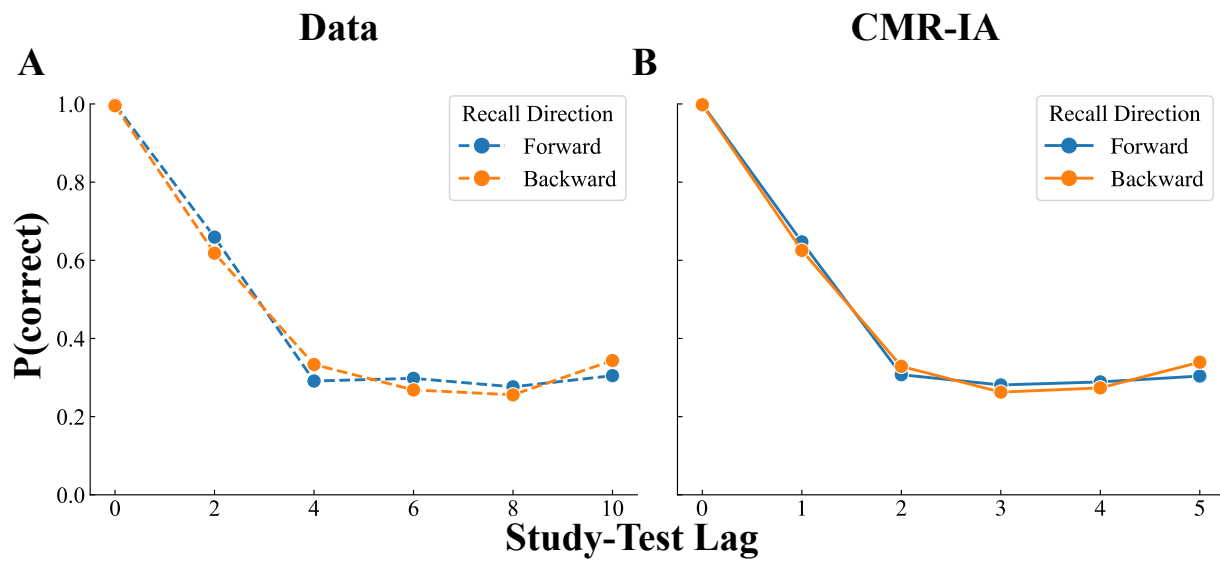
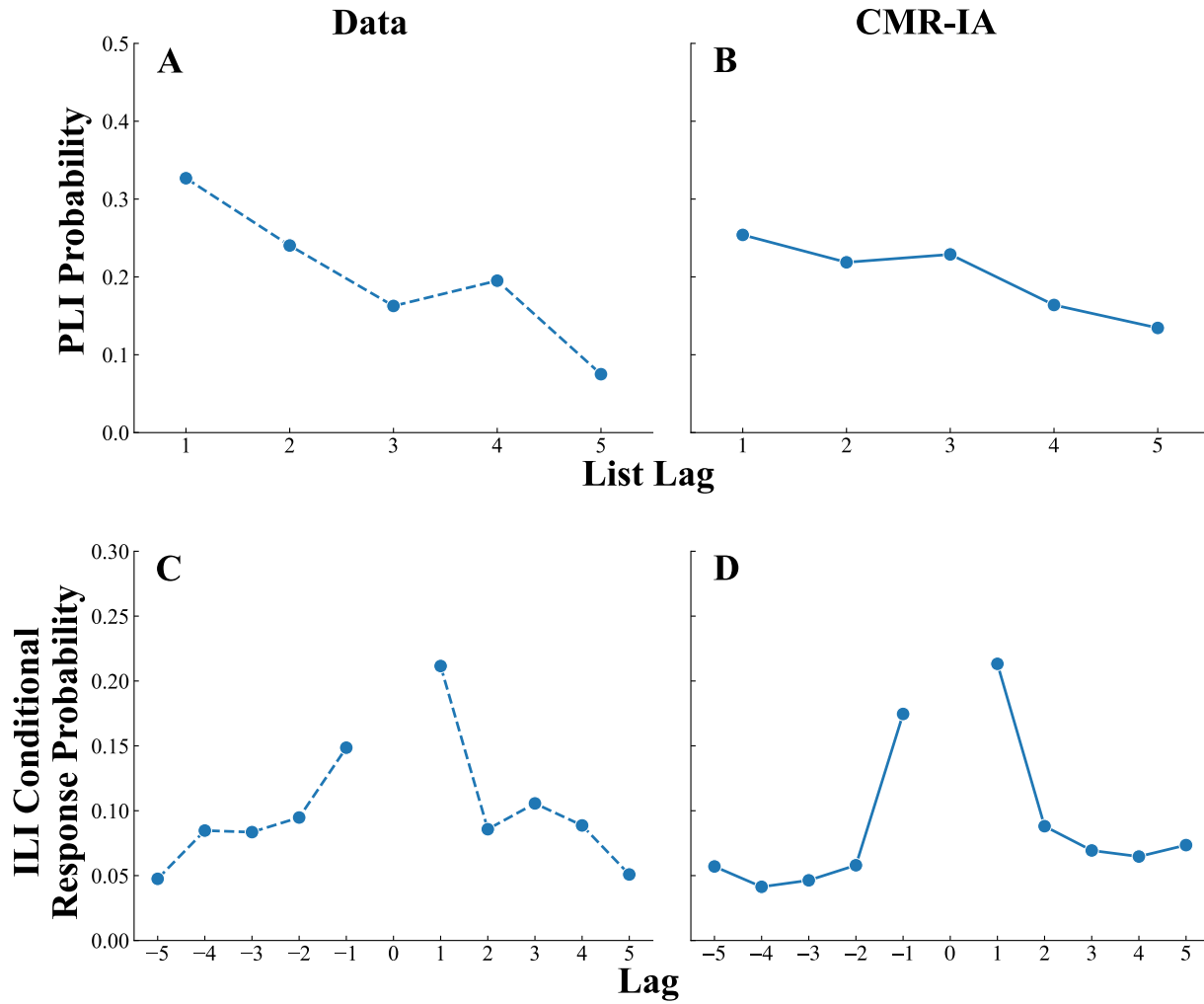


Figure 7

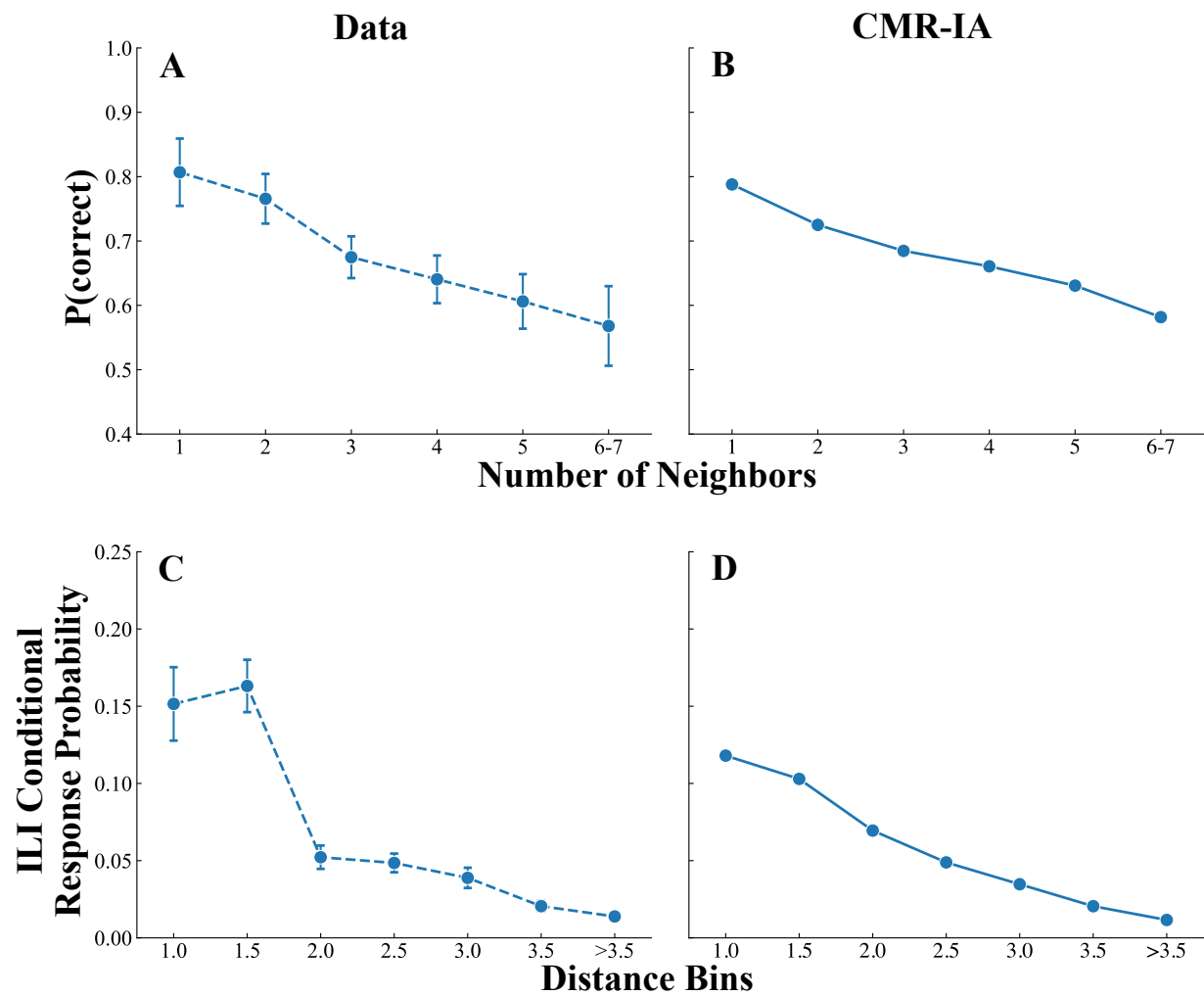
Retention curve in cued recall. Cued recall performance declines as the study-test lag increases in Murdock's (1963) experiment (panel A) and CMR-IA simulation (panel B).

**Figure 8**

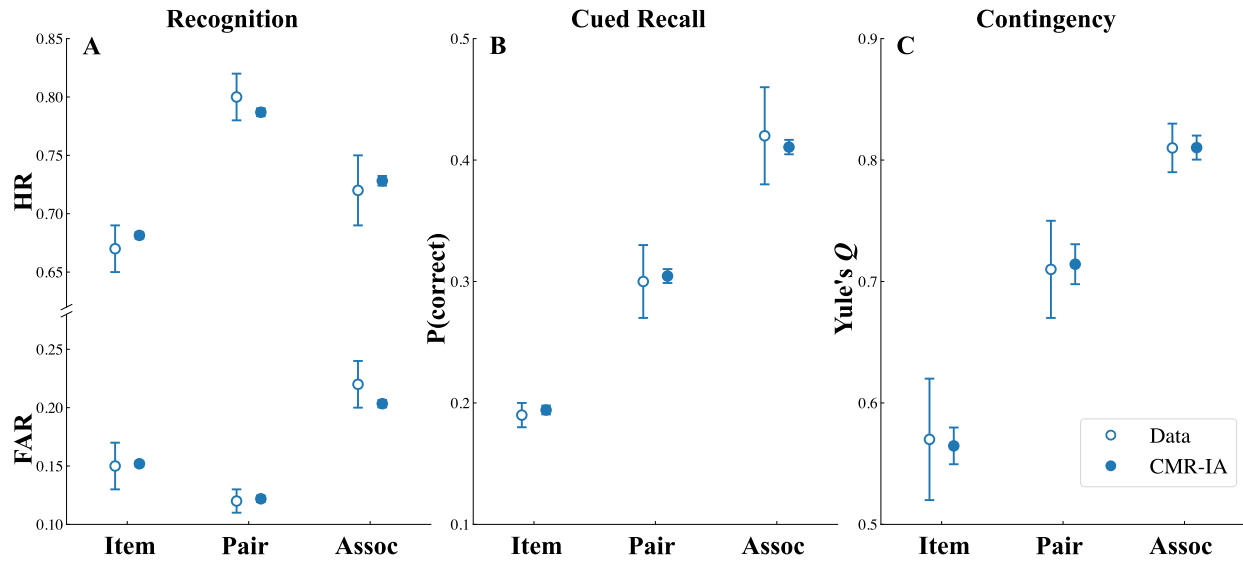
Forward and backward cued recall. Kahana (1993) presented six word pairs to subjects and tested these pairs in reverse order of presentation, replicating an early study by Murdock (1965). The probability of cued recall remains consistent regardless of whether the recall direction is forward or backward (panel A). CMR-IA predicts similar outcomes (panel B).

**Figure 9**

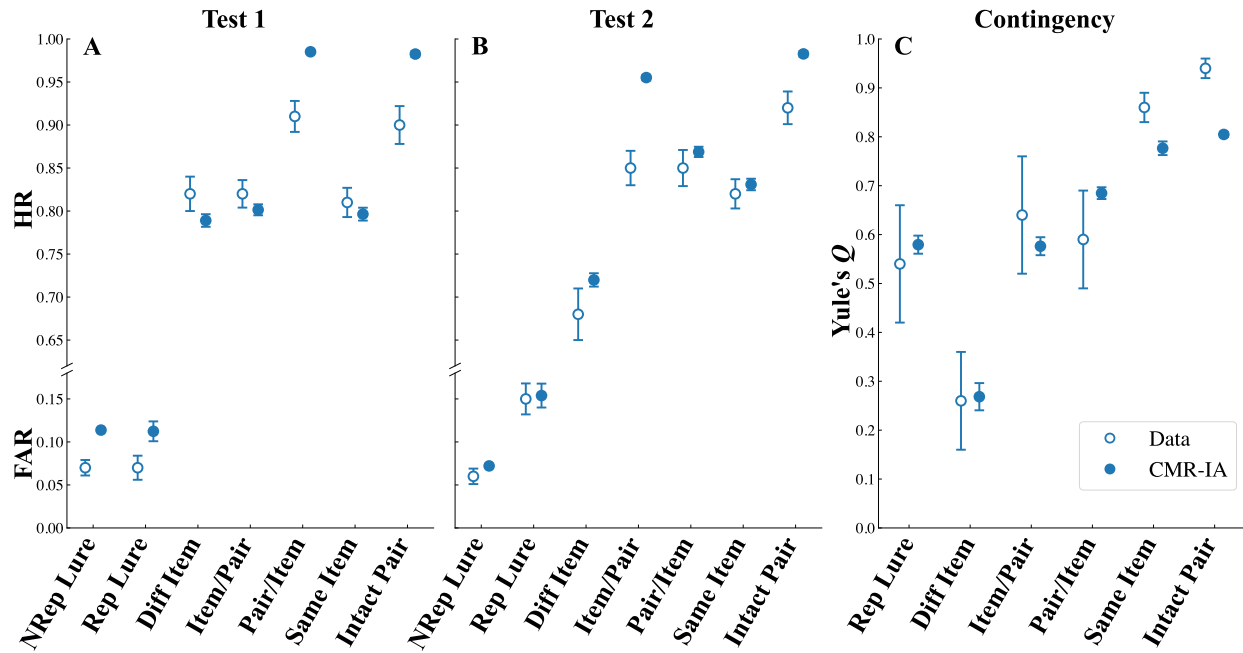
Prior-list intrusions (PLIs) and intra-list intrusions (ILIs) in cued recall. When subjects make a PLI, their incorrect recalls tend to come from recent lists. After studying a list of N pairs ($A_1 - B_1, A_2 - B_2, \dots, A_N - B_N$), an ILI would occur when a subject recalls B_{i+lag} in response to the cue A_i (lag indicates the number of pairs). ILIs tend to arise from pairs studied nearby within the list. The experiment conducted by Davis et al. (2008) illustrated both the PLI recency effect and the ILI contiguity effect (panels A, C). CMR-IA predicts similar outcomes (panels B, D).

**Figure 10**

Similarity effects in cued recall. Pantelis et al. (2008) instructed subjects to learn a list of name-face pairs and subsequently probed them with faces. They demonstrated that the probability of correctly recalling names declines as the number of neighboring faces increases (panel A). Additionally, the conditional probability of making an intra-list intrusion (ILI) decreases with the Euclidean distance between the face corresponding to the incorrectly recalled name and the cue face (panel C). The predictions of CMR-IA align with these findings (panels B, D).

**Figure 11**

Accuracy and Yule's Q values for successive tests: recognition followed by cued recall. Subjects performed either item recognition (Item, A versus X), pair recognition (Pair, $A - B$ versus $X - Y$), or associative recognition (Assoc, $A - B$ versus $A - B_r$) prior to their cued recall test. Panel A displays the observed and predicted recognition results (i.e., hit rate and false-alarm rates for each type of recognition test). Panel B displayed the probability of correct cued recall conditional on the initial recognition test type. Panel C displays the contingency (Yule's Q) between the successive recognition and recall tests. Error bars indicate the standard error of the mean.

**Figure 12**

Accuracy and Yule's Q values for successive recognition tests. Each studied pair undergoes testing twice in Test 1 and Test 2, utilizing either item recognition or pair recognition. We introduced two types of lures during tests: non-repeated lures (NRep Lure) and repeated lures (Rep Lure). Both Experiment 3 and the simulation in CMR-IA indicated that pairs exhibit superior recognition performance in Test 1 (panel A). The same information probed in Test 2 demonstrates an increase in recognition performance (panel B) and a heightened dependency (panel C). Error bars indicate the standard error of the mean.